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From Configurations to Claims:
An Evidence Audit of Urban EV Charging Forecasting

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Abstract

Execution completion is often treated as a proxy for reproducibility, while a statistically positive comparison is often treated as sufficient reason to continue model development. We audit both assumptions in a case study of urban electric-vehicle charging forecasting and extract a reusable claim-control framework linking configuration identity, forecast validity, engineering qualification, and protected-data use. In a frozen inventory of 1,224 UrbanEV configurations, 865 have completed artifacts but only 156 satisfy a joint exact-protocol-and-numerical criterion. A sequential UrbanEV evaluation then shows why later evidence types must remain distinct: strict prequential fusion of a locally audited CAPER expert and compact TimeXer lowers internal macro RMSE by 3.178% relative to the strongest evaluated single model, but increases batch-1 median synchronous latency by 48.86% on the frozen device. Subsequent routing, foundation-teacher, and residual-distillation candidates either deteriorate or remain below their internally time-stamped, project-specific progression gates. On a separate Paris development-only panel, the strict teacher is consistently positive yet its 0.847% point gain remains below the frozen 1% rule. Paris formal and protected target-bearing sources are confined to a procedurally isolated evidence vault and contribute no analytical result. The resulting taxonomy, claim grammar, fail-closed state machine, and artifact ledger show how a forecasting project can preserve partial and negative evidence without promoting it to protected-test, deployment, or state-of-the-art claims.

Keywords: electric-vehicle charging; forecasting; reproducibility; prequential evaluation; experiment audit; latency; stopping rules

1 Introduction

An experiment can finish without reproducing the claim it appears to test. A run may emit a checkpoint and a metric even when the source protocol does not uniquely determine the target, split, preprocessing, or hyperparameters. Conversely, a faithfully specified method may fail for an ordinary engineering reason and never yield a comparable number. Treating these outcomes as one binary label—“reproduced” or “not reproduced”—hides the evidence needed to interpret a forecasting benchmark.

This distinction is consequential in urban electric-vehicle (EV) charging forecasting. Open resources now cover zonal charging demand in Shenzhen, station occupancy in Paris, workplace charging sessions, and harmonized measurements across cities and continents [1–5]. These resources do not define interchangeable prediction problems: occupancy rates, port states, session records, charging duration, and energy volume have different denominators, missingness mechanisms, and operational meanings. At the same time, the associated modeling literature spans physics-informed graph networks, language-model formulations, exogenous-variable Transformers, linear decompositions, and pretrained forecasters [6–11]. A reported score therefore becomes interpretable only after the data semantics, information set, forecast origin, and evaluation protocol are fixed.

Forecasting research has long emphasized out-of-sample and rolling-origin evaluation [12, 13]. More recent guidance documents leakage through preprocessing, inappropriate random splits, weak benchmarks, and metric choices that do not match the data or decision problem [14, 15]. Reproducibility programs likewise encourage code, data, environment, and reporting artifacts [16]. These practices are necessary, but a remaining gap is operational: once a large matrix has been partially recovered and executed, how should a researcher distinguish disclosure quality from execution status, predictive opportunity from realizable gain, and a statistically detectable difference from an improvement large enough to justify added serving cost? Without explicit distinctions, additional model search can gradually convert an evaluation interval into a tuning resource.

We address this gap through an artifact-grounded case audit of one UrbanEV forecasting inventory and its subsequent model-development chain. The intended readers are benchmark authors, artifact evaluators, and reviewers who must decide which urban-forecasting claims are licensed by a heterogeneous experiment record. The audit freezes a six-state configuration taxonomy, verifies target and run identities, and follows each proposed improvement through a sequential evidence ladder. That ladder separates an infeasible oracle diagnostic, forecast-origin predictability, strict prequential fixed fusion, learned routing, synchronous latency, foundation-model teacher qualification, residual distillation, and development-only replication on a separately isolated Paris panel. Fail-closed metric locks prevent incomplete or invalid artifacts from entering aggregate results, while internally time-stamped minimum-effect gates determine whether a stage authorizes the next one. The contribution is an auditable claim grammar and workflow instantiated by this case, not evidence that the framework has already reduced false discoveries across independent teams.

The audit changes the apparent story. Of 1,224 frozen configurations, 865 have completed artifacts, but only 156 satisfy the joint exact-protocol-and-numerical criterion. In the internal UrbanEV rolling evaluation, a strictly fitted CAPER–TimeXer fusion reduces macro RMSE by 3.178% relative to the strongest evaluated single model, yet increases batch-1 median synchronous latency by 48.86% on the frozen laptop-GPU workload. A learned router adds only 0.443% over that fixed reference and therefore fails the internally time-stamped 1% progression gate despite a positive block-bootstrap interval. The same discipline rejects Chronos-2 as the project teacher under the exact UrbanEV protocol, stops residual distillation after a 0.119% gain over ground-truth-only training, and withholds student training on Paris when a strict development-only teacher improves by 0.847% rather than the required 1%. These are scoped gate outcomes, not general verdicts about routing, foundation models, or distillation.

We organize the study around four research questions:

RQ1 How much of a large forecasting configuration matrix is exactly reproducible, approximately recoverable, trend-only, underidentified, failed, or never completed?

RQ2 When two experts are complementary, how much of an infeasible oracle gap survives strict prequential fixed fusion and learned routing?

RQ3 How do apparent accuracy gains change once synchronous inference latency and a frozen minimum-effect threshold are enforced?

RQ4 How do fail-closed audits, control branches, independent evidence, and stopping rules change the claims that remain defensible?

This paper makes four contributions:

- **A configuration-level evidence taxonomy.** We separate protocol and numerical reproduction, disclosure-limited recovery, trend-only recovery, underidentification, execution failure, and non-completion for a frozen 1,224-configuration inventory. The resulting registry makes clear why execution completion is not a substitute for exact reproduction.
- **A portable audit framework and claim grammar.** We type evidence objects as configuration, prediction, target, metric, latency, or protected-storage artifacts; attach identity and lock requirements; and specify the narrow wording each evidence state authorizes. A claim-to-artifact ledger and reviewer checklist make the framework inspectable rather than leaving

it as project history.

- **A sequential audit of model-improvement claims.** We connect target-identity checks, prequential fitting, control branches, block-resampled uncertainty, and hardware-specific latency to explicit authorization gates. This exposes where theoretical complementarity becomes a material fixed-fusion gain and where later routing, teacher, and distillation signals remain below their frozen effect requirements.
- **A fail-closed evidence boundary.** We document audit events that blocked invalid shuffled controls and repaired missing-target representation before metric access, and we procedurally isolate Paris formal and protected target-bearing sources so that only development targets contribute analytically. The workflow produces an auditable mapping from each numerical statement to its protocol, artifact, and permitted wording.

Claim boundary. This is an empirical audit and evaluation study, not a new state-of-the-art model paper. Its numerical conclusions concern one UrbanEV inventory, six internal rolling folds at seed 42, one frozen latency platform, and Paris development-only evidence. UrbanEV and Paris absolute RMSE values are not compared because the panels and targets differ. The 1% threshold is an internally time-stamped project convention, not a universal definition of scientific or economic significance. Neither Paris formal nor Paris protected targets are accessed analytically, and we make no claim of protected-test performance, production readiness, universal generalization, or cross-dataset superiority. Their movement into independent vault-only storage is recorded as a storage event rather than concealed as non-materialization. Within that boundary, the case shows how configuration identifiability and sequential evidence controls constrain the claims this forecasting program can support.

The remainder of the paper first relates this audit to EV forecasting resources, time-series evaluation, and reproducibility research. It then defines the evidence taxonomy and sequential protocol, reports the registry and model-chain results, analyzes the fail-closed events and claim boundaries, and concludes with implications for auditable forecasting studies.

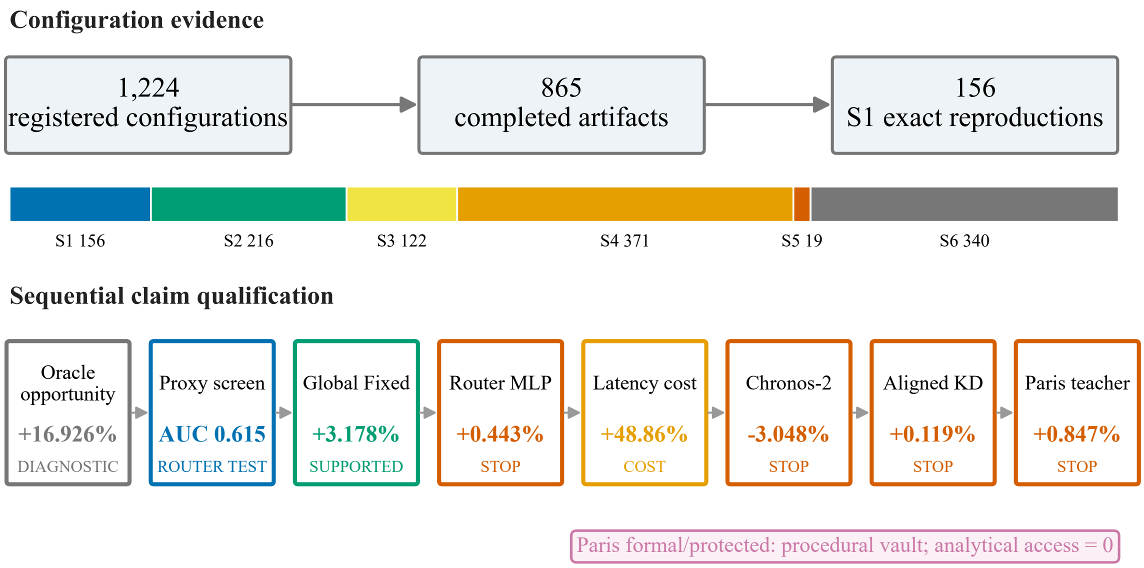

Figure 1: The audit separates registration, completed execution, exact reproduction, predictive opportunity, measured engineering cost, and qualification. A positive diagnostic or comparison does not automatically authorize the next stage; Paris formal and protected targets remain in procedural vault-only storage with zero analytical use.

2 Related Work

EV charging data and prediction. Public EV charging resources differ in both observational unit and prediction target. ACN-Data records workplace charging sessions and supports behavioral and charging-control analyses [4]. UrbanEV instead provides hourly and finer-resolution zonal charging measurements for Shenzhen, together with prices, weather, points of interest, and spatial descriptors [1]. CHARGED harmonizes charging duration, volume, price, weather, and geospatial attributes across six cities, making cross-city heterogeneity a central feature of the resource [5]. The Paris Smarter Mobility dataset records the evolving states of 273 charging points at 91 stations and was designed for occupancy forecasting at station, area, and global aggregation levels [2, 3]. These datasets broaden empirical access, but their targets are not semantically exchangeable. In particular, a session, an hourly charging-duration aggregate, a zonal occupancy ratio, and a non-available-port rate require different denominators and masks. Our study therefore audits target identity and uses each dataset only within its own protocol rather than pooling absolute errors across them.

Target heterogeneity also reflects distinct operational questions. Reviews of open EV load data distinguish session arrivals, connection and charging durations, energy, station load, and occupancy, while the broader forecasting literature addresses infrastructure siting, capacity planning, grid operation, and charging schedules with model-based, probabilistic, and machine-learning approaches [17, 18]. Forecasts may also be required at several aggregation levels and as distributions rather than point estimates: Buzna et al. develop an ensemble for hierarchical probabilistic station-load forecasting that uses information across lower- and higher-level series [19]. This operational lineage motivates treating the target quantity, aggregation level, forecast form, and decision horizon as protocol fields. We use it to delimit valid comparisons, not to claim a new probabilistic or hierarchical forecasting method.

EV forecasting methods have evolved in parallel with these datasets. Physics-informed graph learning incorporates spatial structure and a price–demand prior into regional demand prediction [6], while ChatEV reformulates heterogeneous charging inputs as a text-to-text prediction problem [7]. TimeXer models endogenous temporal patches together with exogenous variables [8]; by contrast, DLinear shows why simple decomposition and linear projection remain necessary controls for long-horizon forecasting claims [9]. We do not attempt to rank these method families across papers. Their reported tasks, data releases, splits, and metrics differ, and our locally audited compact TimeXer and CAPER implementations are project-specific experimental identities. The relevant question for this paper is narrower: which comparisons remain valid after target, history, horizon, fold, and evidence source are made identical?

Time-series evaluation. The prequential view evaluates a sequence of forecasts using information available before each outcome is observed [12]. Rolling-origin evaluation operationalizes this idea by moving the forecast origin through time, optionally refitting the model and producing multiple test periods [13]. Cross-validation for dependent data requires additional care: its validity depends on the prediction setup, stationarity, and residual dependence rather than on a generic rule that all time series must or must not use cross-validation [20]. Empirical comparisons find that temporally ordered out-of-sample estimators are especially appropriate under realistic non-stationarity [21]. Our strict prequential fusion and routing stages follow this lineage: the current evaluation fold never supplies labels for its own fusion weights or router training.

Evaluation design also determines what an error number means. Forecast metrics have different scale, denominator, and target-functional properties, so no single percentage error is reliable in every setting [22]. A broad review of common pitfalls identifies missing simple baselines, leakage through full-series preprocessing, mismatched horizons, and inappropriate aggregation as recurring causes of spurious forecasting conclusions [14]. Forecasting competitions further demonstrate both the value of strong cross-validation and combinations and the fact that complex machine-learning systems do not automatically dominate simpler references [23]. Our audit extends these principles from score computation to stage authorization: a positive difference must retain target identity, beat the strongest eligible reference, satisfy its frozen uncertainty and win-count conditions, and clear a separately declared minimum effect when the claim concerns engineering qualification.

Reproducibility and evidence auditing. Machine-learning reproducibility programs emphasize code, data, environment disclosure, and structured checklists as mechanisms for making empirical results inspectable [16]. Leakage reviews show why inspectability matters: information can cross a nominal train–test boundary through preprocessing, feature construction, temporal misalignment, or repeated analysis even when the final fitting call appears valid [15]. Benchmark results also vary with data sampling, initialization, and hyperparameter selection, and small differences can be dominated by those uncontrolled sources [24]. These findings motivate our run fingerprints, source hashes, target-array checks, and explicit separation between real-ground-truth, proxy, oracle, measured-latency, and artifact-integrity evidence.

Our stopping rules address a related form of adaptivity. A model-selection criterion can itself be overfit, creating optimistic performance estimates even when its underlying estimator is nominally unbiased [25]. Repeatedly querying a holdout also erodes its role as independent evidence; the reusable-holdout literature formalizes this problem for adaptively chosen analyses [26]. We use a procedural rather than privacy-based solution: freeze thresholds before aggregate access, lock metrics until artifacts close, rerun invalid controls non-selectively, and keep later formal and protected intervals physically separate. The contribution is not a new statistical test. It is an artifact-level account of how these principles alter an actual forecasting research program and its permissible claims.

Ensembling, foundation forecasters, and distillation. Forecast combinations often improve accuracy because component models retain different errors, and the M5 competition provides large-scale evidence for both simple and learned combinations [23]. Yet an oracle that selects the better expert after seeing the target is only an upper bound; it does not establish that forecast-origin information can recover the same gain. Our sequence from oracle complementarity to prequential fixed fusion and then learned routing makes that distinction explicit, while synchronous latency tests whether additional inference is justified by the retained accuracy improvement.

Pretrained forecasting models raise a parallel qualification question. Chronos tokenizes scaled numerical series and reports zero-shot probabilistic forecasting across diverse tasks [10]. Chronos-2 expands the formulation to grouped multivariate series and covariates through in-context interaction [11]. Such capabilities motivate direct evaluation, but a model's published aggregate ranking does not determine its role as a teacher under a different target, context length, covariate policy, and point-forecast rule. We therefore treat the locked Chronos-2 checkpoint as a candidate in the exact UrbanEV protocol rather than importing a reported score.

Distillation adds another layer of selection: teacher quality, student capacity, target alignment, and the ground-truth loss all affect whether teacher information improves the deployed student. In this audit, an aligned teacher branch is compared with both an identical ground-truth-only student and a shuffled-teacher control. Beating the shuffled control can identify sample-aligned signal without establishing a material gain over ground-truth training. This distinction connects the distillation experiment to the broader warnings about selection bias and benchmark variance [24, 25]. Accordingly, our related contribution is not a new distillation algorithm; it is a qualification protocol that stops teacher-cache or student expansion when the conjunctive gate fails.

3 Methods

3.1 Audit design and evidence hierarchy

We designed the study as an artifact-grounded audit rather than a search for a new state-of-the-art model. The unit of evidence was therefore not a headline score alone, but a linked record containing the protocol, configuration fingerprint, execution state, prediction and target identities, metric definition, and claim boundary. When records conflicted, we applied a frozen precedence rule: final decision and summary artifacts with an independent audit took precedence over canonical result tables, which in turn took precedence over protocol thresholds and historical progress logs. Existing successful fingerprints were reused; failed or unexecuted configurations could only enter a later study under a new versioned identifier. This rule preserved the execution history and prevented a later successful run from rewriting an earlier failure or non-execution.

The audit separated four evidential questions. *Identifiability* asks whether the disclosed experiment determines a unique protocol and configuration. *Execution* asks whether that configuration reached a terminal run with an artifact. *Predictive evidence* asks whether predictions were evaluated against dataset targets under a time-respecting split. *Engineering qualification* asks whether the resulting effect passed a minimum-effect gate and its accompanying consistency checks. Proxy classification outcomes used to predict which expert would win locally were labeled separately from real-ground-truth forecast metrics and were never treated as forecast-accuracy evidence.

Table 1 is the portable object extracted from the case: each claim must name its evidence object, identity checks, lock or gate, and the strongest wording it authorizes. It is a proposed review instrument derived from one project, not an externally validated guarantee that other teams will reduce false discoveries by adopting it. The complete claim ledger and lock schemas are therefore reported in the reproducibility appendix rather than hidden in an executor narrative.

3.2 Six-level configuration taxonomy

We reclassified a frozen inventory of 1,224 UrbanEV configurations into six mutually exclusive evidence states. The priority order was execution fact (S5 or S6), non-identifiability (S4), and then numerical agreement (S1–S3).

Table 1: Portable fail-closed audit framework. Each transition requires an inspectable artifact; a failed precondition routes to *blocked* rather than to metric interpretation.

Stage	Required state	Minimum portable record	Fail-closed transition	Claim token released
1. Register	protocol fixed before aggregate evidence	data role, target, split, forecast origin, model/config ID, seed, endpoint, controls, gate, stop rule	missing or mutable field $\rightarrow$ `BLOCKED`	`registered` only
2. Execute	registered configuration	run ID, configuration fingerprint, code/environment identity, attempt state, immutable output paths	failure $\rightarrow$ terminal failed run; non-run $\rightarrow$ non-complete	`executed` or `failed`; never `reproduced` by itself
3. Close	all expected cells terminal	expected/observed cardinality; target, order, time, spatial and mask hashes; prediction hashes; provenance links	incomplete or mismatched family $\rightarrow$ repair required	closed-unverified
4. Audit	closed matrix; metrics still locked	identity audit, boundary audit, control validity, repair parent/child hashes, prediction-change flag	any defect $\rightarrow$ invalidate affected formal family; non-selective rerun where selection could bias repair	`audit-pass`
5. Unlock	every registered audit passes	unlock authority, conditions, receipt hash, access counters, exact audited matrix hash	absent receipt or premature access $\rightarrow$ `BLOCKED`	`metric-unlocked` for the named role only
6. Decide	metric implementation and reference fixed	endpoint, reference, effect, uncertainty, consistency checks, conjunctive gate outcome	any required check fails $\rightarrow$ versioned hypothesis terminates	`gate-pass` or `gate-fail`; subchecks stay typed
7. Write	decision and evidence chain retained	claim ID, authoritative artifact and hash, evaluation type, allowed wording, prohibited wording, scope limit	missing lineage $\rightarrow$ claim withheld	scoped manuscript statement

Table 2: Portable fail-closed audit framework (continued): typed claim grammar used to prevent diagnostic, proxy, engineering, or procedural evidence from being promoted to a stronger claim.

Evidence type	Required sentence grammar	Example permitted verb phrase	Promotion that remains prohibited
Oracle diagnostic	[target-dependent role] + [dataset boundary] + [reference] + [opportunity bound]	“indicates an infeasible upper bound”	“is deployable” or “can be recovered”
Proxy prediction	[proxy label] + [forecast-origin information set] + [discrimination/calibration]	“predicts relative expert advantage”	“improves forecast RMSE” or “identifies a cause”
Real-GT development/internal	[data role] + [reference] + [endpoint] + [effect/interval]	“improves internal rolling RMSE by x%”	“generalizes”, “blind test”, or “state of the art”
Engineering measurement	[device + workload + precision + batch] + [endpoint] + [comparison]	“measured P50 was x ms on the frozen workload”	“is efficient” without a portable serving study
Conjunctive gate	[candidate] + [all required checks] + [pass/fail] + [authorized next action]	“failed the 1% point-effect requirement; stage stopped”	replacing the frozen question with a positive subcheck
Artifact-integrity event	[defect] + [pre-metric state] + [repair/audit] + [unlock outcome]	“blocked interpretation until identity audit passed”	“repair improved accuracy”
Procedural data boundary	[asset role] + [storage state] + [analytical-use state] + [threat-model limit]	“vault-materialized; analytical access after lock was zero”	“never materialized” or “cryptographically inaccessible”

S1: protocol and numerical result identifiable and reproduced. The run covered all six folds, followed the official-compatible protocol, and each of the four disclosed metrics differed by at most 2% from its mapped reference.
S2: protocol broadly consistent, with disclosure differences. The run completed and remained above trend-only evidence, but used an audited correction or had a 2–5% numerical discrepancy.
S3: relative trend only. At least one core metric differed by more than 5%, or the disclosed comparison cell could not be covered completely.
S4: original information insufficient for a unique configuration. The paper omitted an identifying element, such as node identity, or an extended configuration could not be mapped uniquely to a disclosed paper cell.
S5: configuration identified but execution failed. The manifest fixed a configuration, but the registry recorded a failed terminal state.
S6: runnable configuration not actually completed. The manifest froze a runnable configuration that was skipped or never registered as a completed run.

S1–S4 can all possess completed artifacts; only S1 denotes exact protocol-and-numerical reproduction under these frozen rules. Classification was a deterministic implementation, not an inter-rater coding study. The necessary/sufficient tests, overlap priority, one configuration example per state, and exact reason fields are given in Section A.1; no alternative tolerance was used to change the frozen counts. Thus, the taxonomy measures the evidence available in this UrbanEV inventory and does not estimate a field-wide reproducibility rate.

3.3 Data boundaries

UrbanEV internal rolling boundary. In this paper's audited UrbanEV task, the target is hourly zonal occupancy rate rather than every quantity in the broader dataset [1]. Occupied-port counts were divided by the frozen zone capacity; the resulting 4,344 × 275 panel was finite and required no target imputation. All same-protocol forecasting comparisons used a 168-hour history, horizons $\mathcal{H} = \{3, 6, 9, 12\}$ hours, six canonical expanding-month folds, and seed 42. Each model in a fold–horizon cell used the same target indices, timestamps, target values, and zone ordering. Fold target ranges, counts, prediction hashes, and fit sources are indexed in Section A.9. This boundary is an internal rolling evaluation, not a blind protected test.

Independent Paris boundary. The second boundary used the Paris Smarter Mobility Data Challenge records [2, 3]. For station z and hour t, the development target was the *non-available-port fraction*,

$$y_{t,z} = \frac{C_{t,z} + P_{t,z} + O_{t,z}}{A_{t,z} + C_{t,z} + P_{t,z} + O_{t,z}}, \tag{1}$$

where A, C, P, and O denote Available, Charging, Passive, and Other counts. Passive and Other were grouped with Charging because the operational target asks whether a port is unavailable for a new user; it is not an energy-demand, session-volume, or pure actively-charging target. Fifty stations with a stable development denominator were locked, totaling 150 ports. The development matrix spanned 3,624 hourly timestamps from 3 July through 30 November 2020; 180,073 of 181,200 station–hour cells were observed (99.378%).

We split the remaining timeline into a 50-day formal role (1 December 2020–19 January 2021) and a 50-day protected role (20 January–10 March 2021). Formal- and protected-role target-bearing sources were storage-materialized only in a procedurally isolated vault outside the main project. Their file identities, row boundaries, and hashes were audited as storage metadata, while analytical access remained zero. Here, *analytical access* means reading a formal/protected target for metric computation, model selection, feature construction, statistical summary, or claim formation; hashing, moving, and inventorying bytes are storage operations and do not increment that counter. Two example rows were displayed during pre-lock schema discovery, before a model or selection rule existed, and are disclosed as a boundary limitation. No formal or protected target was used after the lock for a metric, model decision, or paper result. The separate storage migration is recorded as a storage event and is not described as absence of materialization.

Paris development models shared the same station order, targets, observation mask, history length, and horizons. Within each fold, missing inputs were filled causally: forward filling was limited to four hours, followed by a station-by-source-local-hour-of-week median fitted on the training interval, and then a training-station median. The observation mask was not a model feature, and training loss and evaluation used observed targets only. Because released timestamps did not encode UTC offsets, we used the unique source-naive ordinal-hour sequence; 24- and 168-hour lags therefore refer to ordinal data hours, not reconstructed UTC time.

3.4 Forecast metrics and strict prequential estimation

Let $\mathcal{O}_{f,h}$ be the observed target positions for fold f and horizon h, including all valid timestamps and spatial units. For method m, cell RMSE was

$$R_{f,h}^{(m)} = \sqrt{\frac{1}{|\mathcal{O}_{f,h}|} \sum_{(t,z)\in\mathcal{O}_{f,h}} \left(\widehat{y}_{f,h,t,z}^{(m)} - y_{f,h,t,z}\right)^2}. \tag{2}$$

The primary endpoint was the equal-weight mean of cell RMSEs,

$$\overline{R}^{(m)} = \frac{1}{|\mathcal{F}||\mathcal{H}|} \sum_{f\in\mathcal{F}} \sum_{h\in\mathcal{H}} R_{f,h}^{(m)}. \tag{3}$$

This cell-macro endpoint prevents a fold or horizon with more observed entries from dominating the paper's principal comparison. MAE, WAPE, sMAPE, and RAE were retained as secondary diagnostics. For a candidate c and frozen reference r, positive relative gain denotes lower RMSE:

$$G(c;r) = 100\frac{\overline{R}^{(r)} - \overline{R}^{(c)}}{\overline{R}^{(r)}}. \tag{4}$$

Absolute UrbanEV and Paris RMSE values were not compared because the datasets differ in spatial unit, target construction, missingness, and fold definition.

We used strict prequential fitting [12, 13]: a parameter evaluated in fold f could only use validation data that preceded that fold or out-of-fold predictions from earlier folds. For the two UrbanEV experts, CAPER (C) and compact TimeXer (T), a fixed forecast was

$$\widehat{y}_{f,h}^{\text{fixed}} = \alpha_{f,h}\widehat{y}_{f,h}^{T} + (1-\alpha_{f,h})\widehat{y}_{f,h}^{C}, \qquad 0 \le \alpha_{f,h} \le 1. \tag{5}$$

The Global Fixed variant shared one α_f across horizons, whereas the Horizon Fixed control used four $\alpha_{f,h}$. Fold 1 used only pre-test validation predictions; folds 2–6 used only earlier-fold out-of-fold predictions. The target-dependent oracle selected the expert with smaller pointwise error and served only as an infeasible opportunity bound.

The expert-win classifier used 21 forecast-origin features whose timestamps did not exceed the forecast origin. Six described the experts (CAPER prediction, TimeXer prediction, signed and absolute disagreement, expert mean, and normalized horizon). Fifteen described the observed state: current occupancy, one-step change and its magnitude, daily and weekly deviations and their magnitudes, 24-hour mean, standard deviation and turnover, normalized capacity, and sine/cosine encodings of hour and weekday. Its AUC and Brier score quantified local expert-advantage predictability, not forecast accuracy or a causal mechanism. Learned routing was evaluated separately using real target RMSE. We compared a hard gate, a raw-probability soft gate, and a direct-loss multilayer perceptron; the direct-loss MLP was named as the primary learned-router comparison in the locked router protocol before aggregate Gate 4 access, even though the later raw-soft point estimate was numerically lower.

Table 3: Model identity, local adaptation, and claim scope. "Unknown/not claimed" is used when the frozen project artifacts do not establish a unique upstream identity or an authoritative field.

Paper label	Frozen implementation identity	Data/input contract and local changes	Role and explicit non-claim
UrbanEV CAPER	Existing exact-target expert predictions; legacy artifact label `phase_only`. The latency artifact reports 47,992 parameters. Upstream paper/repository checkpoint identity: **unknown/not claimed**.	UrbanEV panel, 168-hour project context and four horizons in the shared evaluation. The paper reuses predictions and does not reconstruct an unrecorded upstream training recipe or covariate contract.	Single expert, fixed-fusion component, router expert, and KD anchor. Not claimed as an exact replay of an external CAPER release or as a generally defined architecture.
Compact TimeXer-168	Local compact TimeXer-family artifact; 2,358,788 parameters in the latency serving artifact. Exact upstream repository revision for the UrbanEV artifact: **unknown/not claimed**.	Existing UrbanEV predictions with 168-hour context and shared target identity. No new expert training was performed for the audit chain.	Best evaluated single UrbanEV model within the frozen candidate set. Not "official upstream TimeXer", universal best, or SOTA.
Global Fixed	Deterministic convex combination $\alpha_f \widehat{y}^T + (1-\alpha_f)\widehat{y}^C$; one fold-level scalar pooled across horizons. Serving count 2,406,781 parameters includes both experts and scalar.	Fold 1 uses paired pre-test validation; folds 2–6 use earlier-fold OOF predictions only. Current-fold targets cannot fit α_f.	Selected fixed reference and distillation teacher. Not a cost-free ensemble, blind-test result, or cross-dataset claim.
Router v1	Local `CAPER_TimeXer_ReliabilityGate`; direct-loss MLP has hidden widths 32/16, horizon embedding 4, dropout 0.1, Huber loss, seed 42. Authoritative deployed parameter count: **unknown/not claimed**.	Expert outputs/disagreement plus 21 forecast-origin state features, rolling cross-fit; no feature timestamp exceeds the forecast origin.	Frozen learned-router comparison. It failed the 1% gate; no robustness or deployment qualification is claimed.
Chronos-2	Official `amazon/chronos-2` snapshot resolved to commit `29ec3766d36d...e8498c`; `chronos-forecasting 2.2.2`, Torch 2.5.1+cu121. Parameter count: **unknown/not claimed**.	Zero-shot FP32; 168-hour multivariate context; native Q0.5 output; no future covariates; predictions clipped to $[0, 1]$ for the frozen primary metric.	One teacher-qualification screen. Its failure is not evidence that foundation models generally fail for EV forecasting.
Residual student / aligned KD	Local CAPER-anchor plus shared DLinear residual branch; 338 trainable parameters. GT-only, aligned-KD, and shuffled-teacher variants share architecture, initialization, optimizer, data, epochs, and seed.	Occupancy-rate-only input, 168-hour lookback, 275 channels; aligned loss adds the frozen teacher-residual term with $\lambda = 0.5$.	Negative-gated distillation experiment. The aligned-vs-shuffled subcheck is not a passed improvement over GT-only and does not authorize latency or architecture search.
Paris CAPER	`Paris_CAPER_phase_only`, newly trained local adaptation; no UrbanEV checkpoint or fitted coefficient loaded. Authoritative parameter count: **unknown/not claimed**.	Development-only. Available, Charging, Passive, and Other map to available/active/unavailable; daily/weekly ordinal lags; target-history encoder plus station embedding replaces unavailable UrbanEV channels.	Paris single-model candidate. Must not be named unqualified CAPER or treated as an exact UrbanEV replay.
Paris TimeXer	`TimeXer_local_audited_compact_L168`; local source SHA-256 begins `563d4ce4304f`; model class source matches the workspace copy. Upstream commit is **unknown/not claimed**.	Retrained from initialization on Paris development; compact profile ($d = 128$, four heads, two layers); external train-only scaler, masked loss, direct endpoint, no time markers.	Paris single-model candidate. Explicitly a local audited adaptation, not an exact official upstream implementation.
Paris strict teacher	Fold-specific convex combination of the two Paris local adaptations; no separately trained joint model and no protected-target input.	Fold 1 α from pre-evaluation validation; later folds from prior-fold OOF predictions; shared station, mask, target, and timestamp identities.	Development-only teacher qualification. It failed the point-effect gate; student training and formal/protected evaluation were not authorized.

3.5 Uncertainty estimates and conjunctive gates

Paired uncertainty estimates resampled target timestamps in contiguous 24-hour blocks while keeping all spatial units at a timestamp together. The router used 1,000 bootstrap replicates. Distillation and Paris teacher qualification used 5,000 replicates. For Paris, the algorithm intersected timestamps across the four horizons within a fold, partitioned the ordered intersection into consecutive blocks, drew the original number of blocks independently with equal block probability, concatenated the selected timestamps, recomputed every cell RMSE, and then recomputed the equal-cell macro gain while reselecting the better single-model reference in that replicate. Each monthly sequence ended with a 12-hour trailing partial block; retaining it produced variable replicate length but equal expected inclusion for each original timestamp. We did not run post-hoc 48- or 168-hour alternatives after seeing the result. Intervals are therefore descriptive, protocol-specific paired 95% bootstrap intervals for relative gain; a positive interval did not override a failed minimum-effect gate.

Every qualification decision was conjunctive. The 1% point rule was an internal progression convention intended to decide whether added search, cache construction, student expansion, or protected evaluation was worth authorizing; it was not derived from charger-operator utility and is not a universal significance threshold. The relevant lock artifacts predate their respective metric unlocks and are identified by SHA-256 prefixes in Appendix A.7; complete hashes are retained in the machine-readable audit receipt. The displayed prefixes are router protocol `0b1fc56d...e7f18d35`, distillation protocol `fcd07f18...dfe383bc`, and Paris gate specification `96babeb1...90e9561`. The UrbanEV router had to beat TimeXer and both fixed controls, improve macro RMSE by at least 1%, win at least five of six folds and 18 of 24 cells, have a paired interval lower bound above zero, and avoid more than 1% deterioration at any horizon. Chronos-2 teacher qualification used the same 1%, fold, cell, and horizon criteria, together with target identity, causal input exposure, revision reproducibility, and cache feasibility. Residual distillation compared an identical 338-parameter student trained with ground truth only, aligned teacher residuals, or a 24-hour-block shuffled teacher. Passing required aligned KD to improve over GT-only by at least 1%, win five of six folds and 18 of 24 cells, have a positive paired lower bound, preserve every horizon within 1%, and beat the shuffled control in macro RMSE, at least five folds, and paired interval.

Paris teacher qualification was development-only. The strict prequential CAPER–TimeXer teacher had to improve over the replicate-wise best single model by at least 1%, win at least three of four folds and 12 of 16 fold–horizon cells, have a paired lower bound above zero, avoid more than 1% deterioration at any horizon, and pass target, station, mask, timestamp, and origin identity checks. A reported value below 1.000% failed the point-effect gate without rounding or threshold revision.

3.6 Teacher, distillation, and latency protocols

We screened the official `amazon/chronos-2` model [11] from a locked local snapshot at commit `29ec3766...e8498c`; the complete revision is retained in the model-lock artifact. The screen used `chronos-forecasting 2.2.2`, zero-shot FP32 inference, a 168-hour multivariate context, no future covariates, and the native median (Q0.5) forecast. Predictions were clipped to $[0, 1]$ for the frozen primary UrbanEV RMSE, matching the existing occupancy-rate evaluation.

The residual-distillation teacher was the audited Global Fixed fusion. The student combined a CAPER anchor with a shared DLinear residual branch [9]. GT-only, aligned-KD, and shuffled-teacher branches shared architecture, initialization, optimizer, epochs, data, and seed; only the teacher-residual term differed. This negative control tested whether any benefit required sample-aligned teacher information rather than merely an additional regularizer.

Synchronous latency used one RTX 4060 Laptop GPU in FP32 and a single-device sequential execution graph. One request represented a citywide refresh producing all four horizons for all 275 UrbanEV zones. Each method–batch combination used 50 warm-up iterations and

1,000 timed iterations in each of five independent processes; batch sizes were 1, 8, and 32 origins. Model-only time used CUDA events whose terminal event was synchronized. End-to-end wall-clock timing synchronized the GPU before stopping and included input preparation, transfers, four-horizon inference, fusion, and output transfer, but excluded model loading and checkpoint I/O, corresponding to a warm resident service rather than cold start. Batch-1 end-to-end median latency was the frozen primary engineering endpoint; the complete batch 1/8/32 table and software/hardware manifest are reported in Appendix A.8.

3.7 Fail-closed artifact audit

The pipeline withheld evaluation metrics until each registered matrix closed and its predictions passed identity checks. An invalid control mapping, target mismatch, boundary violation, incomplete provenance, or representation defect blocked metric unlock. Corrections made under the lock were non-selective: a defective control family was rerun in full rather than only where an error was observed. Repair receipts recorded before/after hashes and whether predictions changed. Independent reviews were separate high-order read-only checks against a frozen deployment contract; they could block execution or metric unlock but could not edit predictions or lower thresholds. Canonical decision files were retained alongside negative results. Appendix A.3 formalizes the states, required lock fields, unlock conditions, and the weaker receipt-chain evidence that remains a limitation. Finally, the Paris storage guard separated storage materialization from analytical use: formal and protected target-bearing bytes remained in vault-only procedural storage, the main project contained no matching target source, and analytical access count remained zero.

4 Results

4.1 Completed execution was much broader than exact reproduction

The frozen registry contained 1,224 unique configurations with no duplicate run identifier or configuration fingerprint. Of these, 865 (70.67%) had completed artifacts, 19 had failed terminal states, and 340 had not been executed. Completion did not imply exact reproduction: only 156 configurations (12.75%) satisfied S1, while 216 (17.65%) were S2, 122 (9.97%) were trend-only S3, and 371 (30.31%) were non-identifiable S4. Thus, the S1 count was less than one fifth of the completed-artifact count.

The distribution also depended on the original UrbanEV source table. UrbanEV source Table 3 contained all 156 S1 configurations, alongside 180 S2, 94 S3, six S5, and 44 S6 configurations. In contrast, 215 of 216 source Table 4 configurations were S4 under the frozen mapping rule. Source Table 5 contained 156 S4 and 296 S6 configurations among 528 entries. These patterns identify where disclosure and execution evidence was limited; they do not rank the predictive quality of the models appearing in those source tables.

4.2 Strong complementarity contracted under feasible fusion

The lightweight qualification matrix completed all 72 registered runs (three models, six folds, and four horizons). Among the evaluated single models, compact TimeXer had the lowest macro RMSE at 0.073709. DLinear reached 0.075362, NLinear 0.075711, and GBDT-Forecaster 0.078165; CAPER's existing same-target result was 0.075507. These comparisons qualify TimeXer only within the evaluated UrbanEV protocol, not as a universal strongest model.

CAPER and TimeXer nevertheless exhibited substantial local complementarity. Their mean residual Pearson and Spearman correlations were 0.8422 and 0.7638, respectively, and a target-dependent oracle reduced macro RMSE to 0.061233. This was a 16.926% improvement over TimeXer, but it was an infeasible upper bound because it used the realized target to choose the expert. The strict Global Fixed fusion recovered a smaller observed share of this opportunity: macro RMSE was 0.071367, 3.178% below TimeXer. It improved all six fold means and all 24

Table 4: Six-level evidence taxonomy in the frozen UrbanEV configuration inventory. Percentages use all 1,224 registered configurations as the denominator.

Code	Evidence status	Count	Share
S1	Protocol and numerical result identifiable and reproduced	156	12.75%
S2	Protocol broadly consistent; disclosure differences remain	216	17.65%
S3	Relative trend only	122	9.97%
S4	Original information cannot identify a unique configuration	371	30.31%
S5	Configuration identified; execution failed	19	1.55%
S6	Runnable configuration not actually completed	340	27.78%

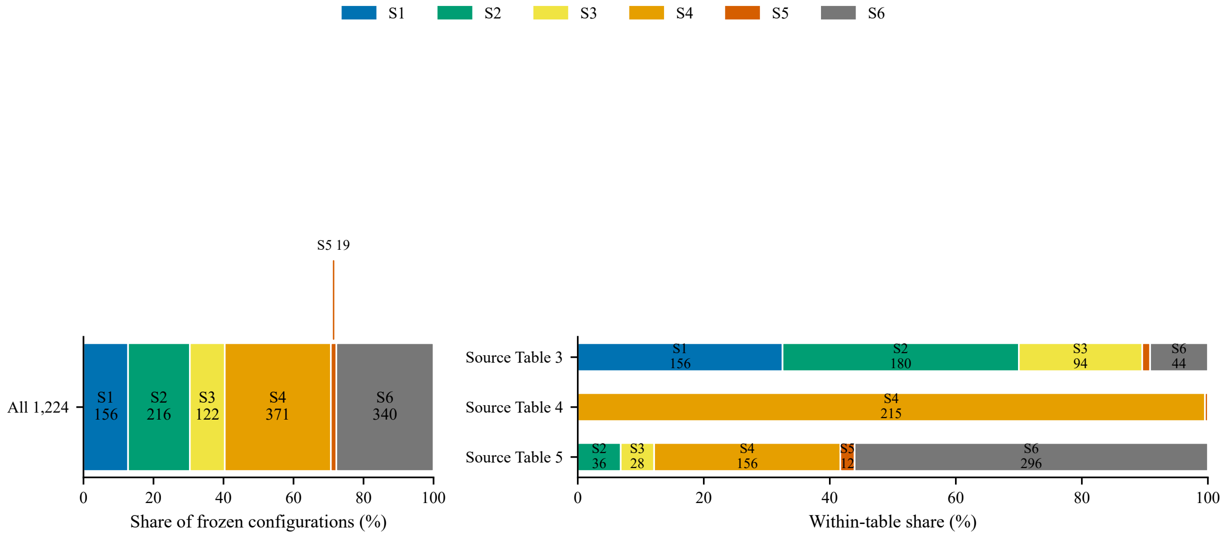

Figure 2: Evidence status in the frozen UrbanEV inventory, overall and by source table. Completed execution and exact reproduction are separate quantities; status counts do not measure intrinsic model quality.

fold–horizon cells; horizon-level gains ranged from 2.890% to 3.645%. A paired block interval was not part of this earlier frozen accuracy-result protocol, so we do not present the result as having passed the later router/teacher gate. The Horizon Fixed control was nearly identical at 0.071372, confirming that post-hoc horizon-specific weights were not needed to obtain the fixed-fusion point result.

Predicting which expert would win was harder than the oracle gap suggested. A rolling classifier using expert outputs and disagreement alone had mean AUC 0.5370 (range 0.5227–0.5701). Adding 21 forecast-origin features increased LightGBM mean AUC to 0.6151, with minimum and maximum fold AUC of 0.5660 and 0.6682; mean Brier score was 0.2388. These proxy diagnostics justified a minimal router evaluation, but did not themselves establish a forecast improvement.

4.3 The learned router produced a positive but sub-threshold gain

The hard router underperformed Global Fixed, with macro RMSE 0.072759. Raw-probability soft routing obtained the lowest observed router RMSE, 0.070979, corresponding to a 0.543% gain over Global Fixed. The locked primary direct-loss MLP obtained RMSE 0.071050, a 0.443% gain; the better observed secondary point estimate did not replace that primary comparison. The MLP won five of six folds and 18 of 24 cells, improved every horizon, and had a paired 24-hour-block gain interval of [0.275%, 0.634%]. Its oracle-gap capture was only 3.123%.

All consistency checks therefore passed, but the 0.443% point gain did not meet the frozen

Table 5: Same-protocol UrbanEV accuracy results over six rolling folds and four horizons (24 equal-weight cells, seed 42). The oracle uses target-dependent expert choice and is not deployable.

Method	Cells	Macro RMSE	Macro MAE	Evidence role
GBDT-Forecaster	24	0.078165	0.049388	lightweight baseline
NLinear	24	0.075711	0.047894	lightweight baseline
CAPER	24	0.075507	0.048480	expert
DLinear	24	0.075362	0.048041	lightweight baseline
TimeXer	24	0.073709	0.046367	best evaluated single model
Global Fixed	24	0.071367	0.044949	strict prequential fixed fusion
Horizon Fixed	24	0.071372	0.044951	horizon-specific fixed control
Oracle	24	0.061233	0.035023	target-dependent upper-bound diagnostic

1% minimum effect. Gate 4 failed and router development stopped. The conclusion is not that routing lacked any signal; it is that the observed internal gain did not justify the added routing complexity under the registered rule.

4.4 The best UrbanEV accuracy carried a clear latency cost

The synchronous benchmark completed 90 of 90 process jobs and retained 90,000 raw timing rows. At batch one, TimeXer required 14.067 ms median and 15.952 ms P95 end-to-end latency. Global Fixed required 20.940 ms median and 33.123 ms P95. Thus, the 3.178% internal RMSE improvement came with 48.86% higher P50 latency and substantially heavier tail latency. NLinear and DLinear were much faster (0.674 and 1.166 ms P50) but had higher RMSE. Under this single-GPU sequential citywide-refresh workload, Global Fixed was the accuracy leader, not a low-cost ensemble. Batch 8/32, model-only, throughput, environment, and portability details appear in Appendix A.8.

4.5 Chronos-2 did not qualify as the UrbanEV teacher

The locked Chronos-2 screen completed all 24 fold–horizon cells and 5,960 forecast origins under the zero-shot, no-future-covariate, median point-forecast setting. Target, index, timestamp, and zone identities passed, all inputs ended no later than the forecast origin, and a repeated sentinel prediction had maximum absolute difference zero. Chronos-2 macro RMSE was 0.073542, compared with 0.071367 for Global Fixed, a relative gain of -3.048%. It beat the control in one of six folds and four of 24 cells; its RMSE deteriorated by 2.138%–3.932% across the four horizons. Although Chronos-2 had lower secondary MAE (0.042250 versus 0.044949), the prespecified decision hierarchy used RMSE as the primary teacher endpoint and MAE as a secondary diagnostic. Chronos-2 therefore failed this exact teacher screen, and Global Fixed remained the audited teacher; this is not a general comparison of foundation forecasters with local models.

4.6 Aligned residual distillation exposed a weak teacher signal, not a stable gain

The final distillation matrix contained 72 audited cells: GT-only, aligned-KD, and shuffled-teacher branches across six folds and four horizons. Macro RMSE was 0.075568 for GT-only, 0.075478 for aligned KD, and 0.075674 for the shuffled control. Relative to GT-only, aligned KD improved RMSE by 0.119%, won four of six folds and 14 of 24 cells, and had a paired interval of [-0.064%, 0.303%]. It failed the frozen gain, fold, cell, and positive-lower-bound criteria, while horizon protection passed.

Aligned KD did outperform shuffled-teacher training by 0.259%, with five of six fold wins and a paired interval of [0.151%, 0.367%]. This contrast supports a small sample-aligned teacher signal,

Table 6: Batch-1 end-to-end accuracy–latency trade-off on the frozen RTX 4060 Laptop GPU, FP32 single-device sequential workload.

Method	RMSE	P50 ms	P95 ms	Origins/s
NLinear	0.075711	0.674	1.056	1483.7
DLinear	0.075362	1.166	1.763	857.9
GBDT-Forecaster	0.078165	6.242	7.195	160.2
CAPER	0.075507	7.934	10.798	126.0
TimeXer	0.073709	14.067	15.952	71.1
Global Fixed CAPER+TimeXer	0.071367	20.940	33.123	47.8

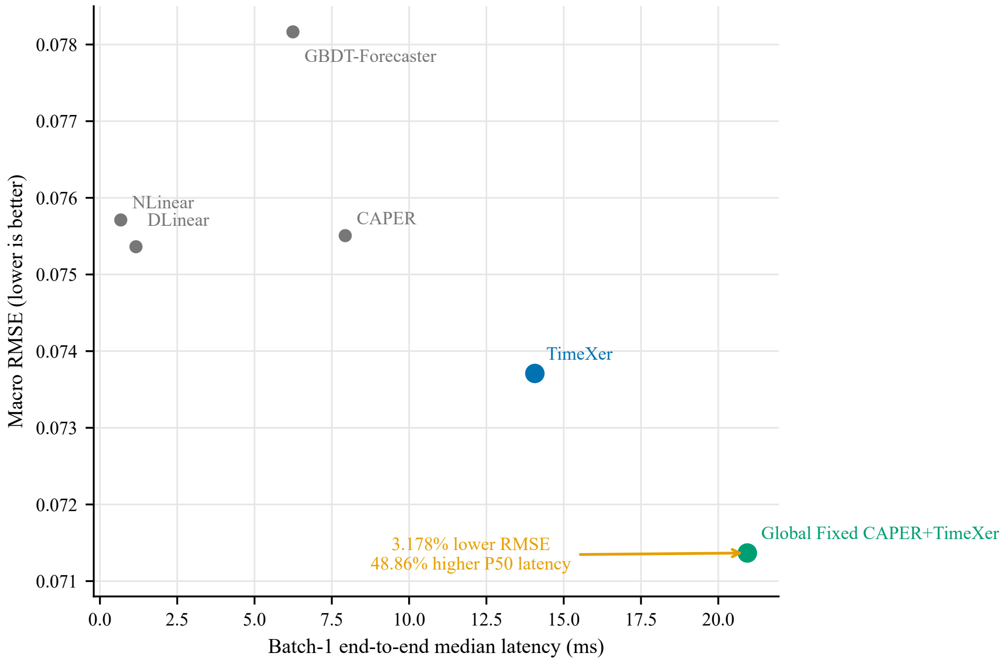

Figure 3: Batch-1 UrbanEV accuracy–latency frontier on the frozen RTX 4060 Laptop GPU in FP32. The arrow from TimeXer to Global Fixed represents 3.178% lower RMSE and 48.86% higher median end-to-end latency.

but not stable improvement over the identical GT-only student. The D1 gate failed, `distill_v1` terminated, and neither student latency nor multi-seed or protected evaluation was run.

4.7 Paris development repeated the distinction between consistency and qualification

Paris non-available-port-fraction results are reported only within the development boundary and are not compared numerically with UrbanEV. They are not energy-demand or session-volume forecasts. Among simple development baselines, last observation had macro RMSE 0.302630, daily lag 0.331654, and weekly lag 0.381137. In the 16-cell teacher qualification, the locally audited Paris CAPER adaptation obtained RMSE 0.244601 and the locally audited TimeXer adaptation 0.242542. The strict prequential fixed teacher reduced RMSE to 0.240488; the target-dependent oracle diagnostic was 0.217661.

The teacher improved over TimeXer by 0.8469%, won all four folds and all 16 cells, and had a paired interval of [0.6867%, 1.0088%]. Every bootstrap replicate selected TimeXer as the best single reference, and no horizon deteriorated. This is a consistent, near-threshold development result, but the point estimate remained below the frozen 1% progression rule. The conjunctive teacher gate therefore failed and the current Paris distillation hypothesis terminated before student training; it was not rounded into a pass and did not authorize formal/protected access.

Formal- and protected-role target-bearing sources existed only in the independent procedural vault. The storage boundary was materialized and hash-audited, but analytical access count remained zero; no post-lock target value or metric from either role contributed to the Paris result or this paper.

Table 7: Stage-specific qualification outcomes. Gains use different frozen references and are not absolute cross-stage or cross-dataset comparisons. "–" for Chronos-2 means that its frozen screen did not register a paired interval, not zero uncertainty.

Stage	Frozen reference	Gain	Paired 95% CI	Primary point gate	Decision
Router MLP	Global Fixed	+0.443%	[0.275, 0.634]%	1.0%	FAIL
Chronos-2	Global Fixed	-3.048%	–	1.0%	FAIL
Aligned KD	GT-only student	+0.119%	[-0.064, 0.303]%	1.0%	FAIL
KD vs shuffled	shuffled teacher	+0.259%	[0.151, 0.367]%	positive control contrast	PASS SUBCHECK
Paris teacher	Paris TimeXer	+0.847%	[0.687, 1.009]%	1.0%	FAIL

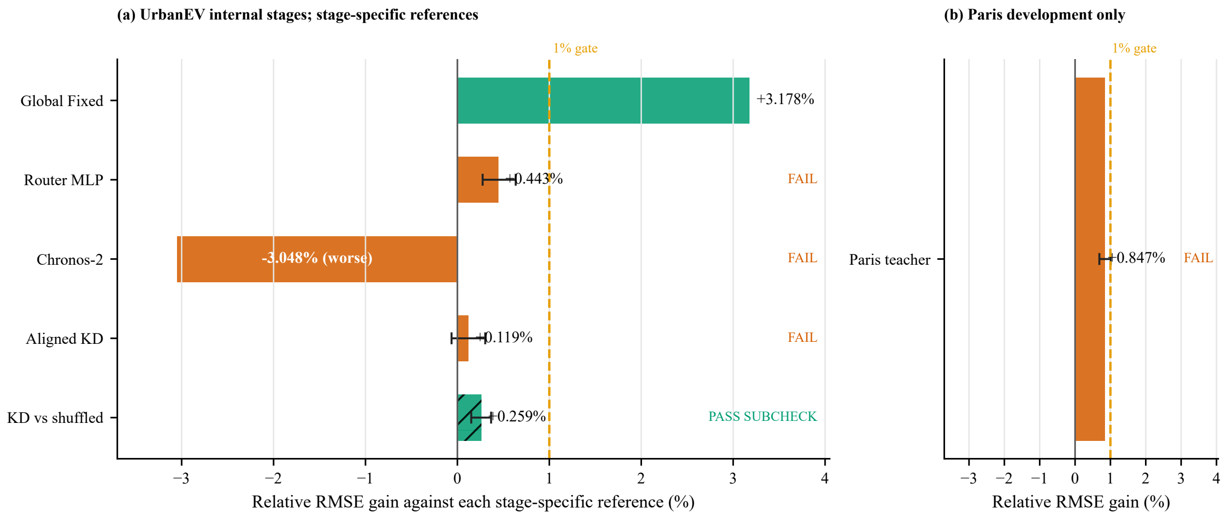

Figure 4: Stage-specific relative gains against each frozen reference. UrbanEV and Paris use different panels, targets, and references; the plot compares gate outcomes rather than absolute RMSE. The dashed 1% line is the project-specific minimum effect, not a universal utility threshold.

4.8 Fail-closed audits changed the admissible evidence before interpretation

The first M7-C matrix audit passed 69 of 72 cells and detected three shuffled-control mappings that were identities. Metrics remained locked. The pipeline invalidated and reran all 24 shuffled controls rather than selecting only the three detected cells; the final audit passed all 72 cells before D1 metrics were computed.

The Paris pipeline was similarly fail-closed. An initial independent review blocked deployment with eight implementation and evidence-boundary requirements. After the development-only shard, fingerprints, metric lock, synchronized bootstrap, provenance, and recovery rules were repaired, a final differential review issued `DEPLOY_PASS`. All 32 model jobs then closed before metric access. A representation audit found fill values at masked target positions in 24 of 32 artifacts, covering 10,236 entries. The repair restored missing-value representation without retraining and left every prediction unchanged; metrics had not been accessed. A subsequent prediction audit passed 32 of 32 artifacts and then unlocked the development metrics.

These interventions support the integrity of the final negative decisions; they do not constitute evidence that auditing improved forecast accuracy. Throughout the Paris stage, formal/protected sources remained in vault-only procedural storage and analytical access count stayed at zero.

Table 8: Fail-closed interventions performed before scientific interpretation. Repairs did not change model predictions or optimize observed metrics.

Event	Detection	Required action	Final state
M7-C control	3 identity mappings detected (69/72 initial audit pass)	all 24 shuffled controls rerun	72/72 final audit PASS
Paris predeployment review	independent review: BLOCK	8 blockers repaired; DEPLOY PASS	pre-run authorization only; metrics locked
Paris target repair	24 runs; 10,236 masked targets restored	restore missing representation; predictions unchanged	metrics still locked
Paris metric unlock	32/32 jobs closed	prediction identity audit	prediction audit PASS; metrics unlocked; teacher gate FAIL; formal/protected analysis locked; analytical access 0

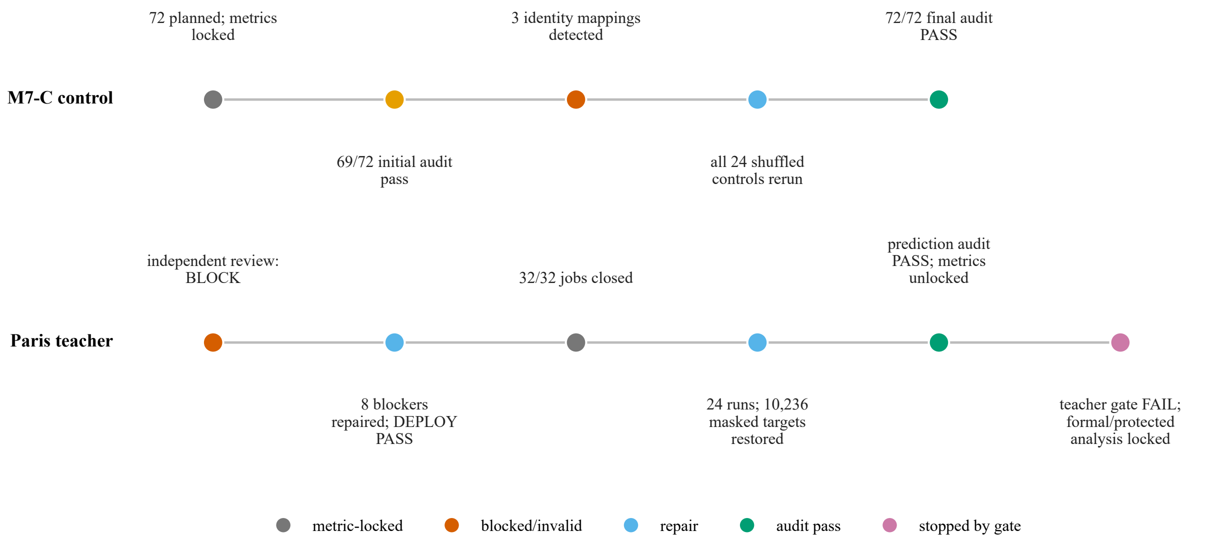

Figure 5: Fail-closed audit timelines. Invalid controls and missing-target representation were corrected before metric interpretation; the corrections did not alter model predictions or authorize formal/protected analysis.

Table 9: Research-question map from operational definition to bounded answer.

RQ	Operational question	Evidence and analysis	Direct result	Bounded answer
RQ1	How many frozen configurations are exact, disclosure-limited, trend-only, underidentified, failed, or non-complete?	1,224-row status registry; deterministic S1–S6 classifier; execution state, source-cell mapping, fold coverage, and worst disclosed-metric grade.	865 artifacts complete; S1=156, S2=216, S3=122, S4=371, S5=19, S6=340.	Completion is broader than exact reproduction in this inventory. Counts are not a field-wide reproducibility estimate or a model ranking.
RQ2	How much oracle complementarity survives feasible prequential fusion and learned routing?	Target-dependent oracle; expert residual diagnostics; forecast-origin proxy classifier; strict Global/Horizon Fixed controls; direct-loss and secondary router variants.	Oracle +16.926% versus TimeXer; Global Fixed +3.178%; primary router +0.443% versus Global Fixed.	A material fixed-fusion gain survives, but only a small additional router signal passes to the real-GT stage and it fails the frozen effect gate.
RQ3	What changes after synchronous latency and the minimum-effect rule are enforced?	RTX 4060 FP32 batch 1/8/32 benchmark; batch-1 primary endpoint; router, Chronos-2, KD, and Paris conjunctive gates.	Global Fixed has +48.86% batch-1 P50 versus TimeXer; later gains are +0.443%, −3.048%, +0.119%, and +0.847%.	The accuracy leader has a measured serving cost; positive or consistent subchecks do not automatically qualify the next engineering stage.
RQ4	How do fail-closed audits and stopping rules alter admissible claims?	Metric lock; non-selective control rerun; 32/32 Paris prediction closure; unchanged-prediction repair; versioned termination; procedural vault guard.	Three identity shuffles were excluded before metrics; 10,236 masked target entries were restored without prediction changes; analytical access after the Paris lock remained zero.	Integrity controls changed what could be interpreted and stopped failed versions. They do not prove higher accuracy, hardened security, or external validity.

5 Discussion

5.1 Completion is not reproduction

The configuration audit exposes a distinction that a conventional completed-run count would hide. Of the 1,224 registered UrbanEV configurations, 865 have completed artifacts, yet only 156 satisfy the frozen S1 criterion: the protocol is identifiable and all disclosed numerical results are reproduced within the prespecified tolerance. The remaining completed artifacts occupy qualitatively different evidence states. Some are broadly consistent but differ from the disclosure (S2), some recover only relative trends (S3), and others cannot be mapped to a unique source configuration (S4). A successful process exit can therefore establish that one implementation ran; it cannot establish that the implementation represents the reported experiment. This distinction also explains why configuration status should not be read as model quality. For example, the concentration of S4 entries in one source table reflects missing identification information, not an empirical finding that the associated models are weak.

This layered view is stricter than a binary "reproduced/not reproduced" label, but it is also more useful. It preserves partial evidence without promoting it to an exact-reproduction claim. The distinction is consistent with reproducibility frameworks that separate artifacts, procedures, and numerical outcomes [16, 24]. In our inventory, the practical consequence is large: reporting 70.67% artifact completion rather than 12.75% S1 reproduction would answer a different research question.

5.2 Oracle opportunity is not deployable gain

CAPER and TimeXer exhibit substantial local complementarity. A target-dependent oracle that selects the lower-error expert for each target reduces RMSE by 16.926% relative to the best evaluated single model. This value is informative as an upper-bound diagnostic, but the oracle uses the realized target and is therefore infeasible at forecast time. Once the decision must be made from information available at or before the forecast origin, the evidence contracts sharply. Strict prequential Global Fixed fusion retains a 3.178% internal RMSE improvement, whereas the learned direct-loss router adds only 0.443% over that fixed reference and recovers 3.123% of the oracle gap.

The gap between 16.926% and 0.443% is not a contradiction. It measures the difference between complementary errors that exist after outcomes are observed and complementary errors that can be exploited prospectively. The state-feature classifier's mean AUC of 0.6151 supports moderate predictability of relative expert advantage, not causal knowledge of charging demand and not equivalent forecast performance. This is why the evidence ladder separates oracle diagnostics, proxy-label prediction, real-ground-truth forecasting, and engineering qualification rather than treating them as interchangeable model scores.

Deployment relevance contracts the result once more. Global Fixed has the best audited UrbanEV internal RMSE, but synchronous batch-one inference on the frozen RTX 4060 Laptop GPU increases end-to-end P50 latency from 14.067 ms for TimeXer to 20.940 ms, a 48.86% overhead; P95 rises from 15.952 to 33.123 ms. The fixed fusion is thus an accuracy result with a measured hardware-specific cost, not a cost-free or universally efficient ensemble. The observed accuracy–latency trade-off also shows why a predictive improvement and a deployment recommendation require different evidence.

5.3 A positive confidence interval is not a frozen qualification decision

Several stages produced positive signals without satisfying their internally time-stamped gates. The router improved on Global Fixed by 0.443%, with a paired 24-hour-block interval of $[0.275\%, 0.634\%]$. The interval is entirely above zero, but it is entirely below the frozen 1% minimum-effect threshold. The Paris strict teacher provides an even sharper example: it won all four development folds and all 16 fold–horizon cells, and its interval was $[0.687\%, 1.009\%]$, yet its point improvement was 0.847%. The conjunctive rule required a point improvement of at least 1%, so the teacher did not qualify. The aligned distillation branch was weaker still: its 0.119% improvement over the identical ground-truth-only student had an interval of $[-0.064\%, 0.303\%]$ and failed the frozen fold and cell requirements.

These decisions do not assert that effects below 1% are scientifically meaningless. The threshold is a local progression convention, not an operator utility model or a universal economic or statistical constant. Rather, the decisions preserve the interpretation fixed before result inspection. A confidence interval evaluated against zero addresses evidence for a signed effect; an engineering gate addresses whether the observed effect is large and consistent enough to justify the next stage. Treating the former as automatic satisfaction of the latter would silently replace the registered question after observing the answer. As post-hoc context only, a 0.5% point rule would still reject the router, Chronos-2, and aligned KD but would change the Paris development decision; a 2% rule would reject every downstream candidate. These counterfactuals do not revise the frozen 1% outcome.

5.4 Stopping rules make negative evidence reusable

The stopping rules prevented small positive results from initiating open-ended searches on the same evidence interval. Router v1 stopped after its effect-size failure. Chronos-2 remained an auditable zero-shot teacher screen rather than motivating revision-specific tuning after its −3.048% gain against Global Fixed. Distillation v1 terminated after the aligned branch failed its main gate, even though it beat the shuffled control by 0.259%. The current Paris teacher hypothesis likewise stopped on development rather than opening formal or protected targets to rescue a near-threshold point estimate. These stops protect future evidence boundaries and make the negative results interpretable: each failed branch answers a frozen question instead of becoming one iteration in an unreported adaptive search [25, 26].

Fail-closed controls reinforced the same discipline at the artifact level. In the distillation matrix, three identity mappings invalidated the intended shuffled controls while metrics remained locked. The audit invalidated and reran all 24 shuffled-control cells non-selectively, then unlocked metrics only after the final 72/72 artifact audit passed. In Paris, an independent predeployment review first blocked execution on eight issues. After correction, the 32 prediction artifacts closed without evaluation metrics; a later representation audit restored 10,236 masked target entries to missing values across 24 runs without changing predictions, and metrics remained inaccessible until all 32 artifacts passed the identity audit. These interventions support integrity of the final evidence. They do not support a claim that repair improved model accuracy.

5.5 The vault is an evidence boundary, not a performance result

Paris formal and protected target-bearing sources now reside outside the main project in a procedurally isolated evidence vault. The corrected inventory contains 48 non-self-referential files, including 29 hidden Git files and the object pack that could otherwise reconstruct removed ground-truth sources. Every manifest entry matches its recorded byte count and SHA-256 hash. Formal and protected storage shards cover 2020-12-01 through 2021-01-19 and 2021-01-20 through 2021-03-10, respectively. Storage materialization is recorded separately from analytical access: one migration event occurred, while the protected analytical-access count remains zero. The failed-attempt partial files are also inventoried as target-bearing evidence and were not analyzed.

This boundary supports a narrow but important claim: no post-lock Paris formal or protected target contributed to model selection, metric computation, or a paper result. It does not imply that the bytes never existed, that two pre-lock schema rows were never visible, or that Paris development is blind-test evidence. The protected assets include raw sources, reference files, Git objects, final shards, and failed partials. In-scope failures are accidental analysis, stale-cache reuse, path leakage, and recoverable Git history; a malicious authenticated user, operating-system compromise, and cryptographic confidentiality are out of scope. The vault is a path, manifest, hash, and process boundary with inherited filesystem permissions, not encrypted storage or hardened access control. Its value is epistemic: it makes the declared absence of analytical use inspectable and preserves those targets for a future versioned protocol.

5.6 Implications for EV charging forecasting studies

The case yields positive guidance rather than only prohibitions. Authors should name whether a target represents energy, sessions, active charging, occupancy, or port non-availability; report denominators and observation masks; separate model-search intervals from future operational evaluation; and tie latency to a concrete request shape. Reviewers should ask whether a completed run is still underidentified, whether an oracle or proxy has been promoted to forecast evidence, and whether a positive interval silently replaced a frozen minimum-effect question. Operational follow-up should complement macro RMSE with stakeholder-specific measures such as peak-period error, high-occupancy recall, calibrated intervals, or denied-service proxies for driver-facing systems. This paper does not estimate those utilities.

The portable author/reviewer checklist is reported in Appendix Table 17 so that it can be reused independently of this case narrative.

6 Limitations

Scope of the registry. The 1,224-configuration taxonomy describes one frozen UrbanEV reproduction inventory. Its S1–S6 assignments depend on the available source disclosure, deterministic mapping rules, and internally fixed 2% and 5% numerical tolerances. The classifications were not independently double-coded, and no inter-rater reliability estimate is available. The resulting 12.75% S1 rate is not an estimate of reproducibility across urban forecasting as a field. Status counts also measure evidence identifiability and execution state, not intrinsic model quality.

Framework validation. The proposed audit framework is extracted from a single self-audited project. Independent high-order reviews challenged specific protocol and result packages, but no external team has applied the full claim grammar, state machine, or S1–S6 decision tree to a separate benchmark. The paper therefore offers a reusable instrument and worked case, not evidence that the instrument improves decisions or reduces false discoveries across laboratories.

Internal and development-only evaluation. The UrbanEV comparisons use six internal rolling folds, four horizons, and seed 42. Paris teacher qualification uses four development folds over a different panel, non-available-port target, missingness mask, and local audited model adaptations. Absolute UrbanEV and Paris RMSE values are therefore not comparable. The windows also differ behaviorally: Shenzhen covers September 2022–February 2023, whereas Paris development covers July–November 2020, when pandemic-era mobility, policy, infrastructure, pricing, land use, and fleet composition could differ. The evidence controls protocol leakage, not those transport-domain shifts. No UrbanEV protected evaluation was opened, and no post-lock Paris formal or protected target was used analytically. The results do not establish blind-test generalization, robustness across seeds or cities, production readiness, or state of the art.

Restricted candidate and implementation set. The baseline conclusion applies only to the evaluated candidates. The Chronos-2 screen fixes one official model revision, zero-shot FP32 inference, and no future covariates; it does not establish that foundation models generally underperform on charging data. Likewise, the CAPER and compact TimeXer Paris implementations are local audited adaptations rather than claims about every upstream implementation. The residual-distillation result concerns one 338-parameter student and one seed, and the stopped protocol did not authorize a student architecture search, multi-seed expansion, or latency evaluation.

Oracle, proxies, and uncertainty. The oracle is target-dependent and only bounds latent complementarity. The expert-advantage classification task uses a proxy label derived from relative expert errors; its AUC does not prove causal predictability or forecast improvement. Paired block-bootstrap intervals account for temporal dependence under the chosen 24-hour blocking scheme, but they do not eliminate all uncertainty from fold construction, model selection, or a small number of seeds. Each Paris fold retained a trailing 12-hour partial block, and no post-hoc 48- or 168-hour sensitivity was run, so its interval should be interpreted under that exact resampling implementation. The 1% minimum-effect gate is a project-specific progression rule without an accompanying economic utility model, and it should not be generalized as a universal significance threshold.

Hardware-specific latency. Latency was measured in FP32 under single-device sequential execution on one RTX 4060 Laptop GPU; batch one was the frozen primary endpoint, while batches 8 and 32 are descriptive appendices. A request produced predictions for 275 zones and four horizons. Kernel fusion, parallel expert execution, quantization, compilation, batching, cold start, a different accelerator, or a serving system could change both the absolute timings and the fusion overhead. The measured 48.86% P50 overhead is reproducible for the frozen warm-resident workload, not portable to arbitrary deployments.

Audit coverage. Fail-closed checks detected identity shuffles, target-representation errors, incomplete provenance, and premature metric access. These checks reduce known integrity risks but cannot prove the absence of every implementation defect. The repairs were verified through hashes, identities, and unchanged predictions; they were not independently reimplemented from scratch. The independent Paris result review retained non-decisive warnings: the teacher-weight CSV did not carry complete upstream provenance, the pre-unlock state lacked a separate hash snapshot, and the receipt-update chain was weaker than the prediction-identity audit itself. None can reverse a point estimate below the frozen gate, but they limit stronger process claims. Two example source rows were displayed during early schema discovery before the Paris boundary lock, when no model or selection rule existed. This event was logged and was not used for model choice, but it prevents an absolute claim that no target value was ever visible before boundary creation.

Dataset-license metadata. The Paris challenge repository and the associated Zenodo record expose different license labels (ODbL and CC BY 4.0, respectively) [2, 3]. This paper treats them as source-specific metadata and does not resolve the legal relationship between the records. Downstream redistribution should therefore verify the applicable terms at the exact source used.

Vault boundary. The final vault manifest covers 48 non-self-referential files, including 29 Git files, and all recorded hashes verify. Analytical access to protected labels is zero, but storage migration materialized formal and protected shards and retained redundant raw, reference, final-shard, and failed-partial copies. The partials were inventoried but not analyzed. The vault inherits filesystem permissions, including modify rights for authenticated users; it is procedural isolation rather than encryption or hardened authorization. Some truth-derived tutorial figures and score summaries remain outside the exact-target isolation set because they cannot reconstruct station-level labels. These properties limit the claim to verified source separation and zero analytical use.

Negative-result interpretation. A failed frozen gate does not prove that an entire method family cannot work. It establishes only that the audited candidate, reference, data interval, and qualification rule did not authorize progression. Conversely, a passed subcheck—such as aligned distillation outperforming a shuffled teacher—does not upgrade the failed main comparison. Our conclusions concern the evidential status of these experiments, not universal impossibility.

7 Conclusion

This audit separated four questions that are often collapsed in large forecasting studies: whether a configuration is identifiable, whether an implementation completed, whether it carries predictive signal, and whether the resulting gain is large and reliable enough to justify progression. In the frozen UrbanEV registry, 865 of 1,224 configurations have completed artifacts, but only 156 meet the exact-protocol-and-numerical S1 criterion. Completion is therefore useful evidence, but it is not reproduction.

The same separation changed the model narrative. A target-dependent oracle exposed 16.926% complementarity, while strict prequential fixed fusion retained a 3.178% internal gain and incurred a 48.86% batch-one P50 latency overhead on the frozen device. Learned routing, Chronos-2 teacher qualification, aligned residual distillation, and the Paris development teacher each answered their registered questions, but none passed its frozen progression gate. In particular, statistically positive router and Paris intervals did not override the project-specific 1% minimum-effect requirement.

Fail-closed metric locks, non-selective control reruns, prediction-identity audits, and internally time-stamped stopping rules kept these negative results interpretable. The procedural Paris vault further separates future formal/protected evidence from the present manuscript: 48 non-self-referential files, including 29 recoverable Git files, are hash-inventoried outside the main project, while protected analytical access remains zero. This is a workflow boundary, not encrypted storage and not a protected-test result.

The central conclusion is methodological rather than architectural. In this case, separating configuration identity, execution, statistical evidence, practical effect, latency, and protected-data use makes the supported scope inspectable. The accompanying claim grammar is a reusable proposal, not yet a cross-team validated instrument. Future work should apply a newly frozen model and loss specification to an untouched boundary, use multiple seeds and hardware environments, and independently test the same fail-closed progression rules. Until such evidence exists, the strongest supported outcome is an auditable internal accuracy–cost result accompanied by explicit negative gates and preserved future evaluation data—not a claim of state of the art or deployment readiness.

A Reproducibility and Claim-Control Appendix

This appendix exports the audit interface behind the case study. It is deliberately documentary: it reuses frozen main-workspace artifacts, performs no model training or inference, recomputes no experiment metric, runs no new bootstrap or threshold sensitivity, and does not read Paris formal or protected target content. The tables distinguish facts present in the artifacts from portable recommendations. When a field cannot be established from a frozen artifact it is reported as *unknown/not claimed*; it is not reconstructed by assumption.

The appendix also does not supply evidence that the study did not collect. In particular, no independent-coder classification exercise, inter-rater agreement statistic, alternative bootstrap block-length analysis, or new Global Fixed uncertainty interval is claimed. The main text's 0.5/1/2% threshold counterfactual is arithmetic context only; it does not recompute metrics or alter a decision. The S1–S6 registry is a deterministic single-pipeline classification of one inventory.

The label “independent review” in the project record refers to a separately generated high-order differential artifact review, not to a blinded external human replication or an inter-rater study.

A.1 Deterministic S1–S6 decision tree

The classifier in `experiments/00_registry_reclassification/scripts/rebuild_status_registry.py` applies the following ordered rules to each frozen manifest row. Order is part of the definition; a later numerical rule cannot overwrite an earlier execution fact.

D1. If `execution_status=failed`, assign S5 and require any future attempt to use a new versioned retry identifier.

D2. Otherwise, if `execution_status` is not `complete`, assign S6 and require any future execution to use a new versioned run identifier.

D3. Otherwise, if the row maps to UrbanEV source Table 4, assign S4 because the source paper and repository do not identify the disclosed node uniquely.

D4. Otherwise, if no source comparison cell can be mapped uniquely, assign S4.

D5. Otherwise, collect the disclosed RMSE, MAPE, RAE, and MAE grades and take the worst valid grade. The frozen bands are exact (relative error no greater than 2%), near (2–5%), and trend (greater than 5%); an empty valid-grade set defaults to trend when a comparison record exists.

D6. If comparison fold coverage is below six, assign S3. Otherwise, if the worst grade is trend, assign S3.

D7. Otherwise, if the track is `official` and the worst grade is exact, assign S1. Every other completed, uniquely mapped, non-trend row is S2.

Thus the executable priority is S5/S6 $\rightarrow$ S4 $\rightarrow$ S1/S2/S3. A completion mark indicates process completion only; it is not used as a synonym for S1. The following examples are the first row of each final status in the authoritative CSV. The `classification_reason` text is an English rendering of the Chinese source field so that the appendix compiles under the paper’s Latin-font setup.

Table 10: One frozen registry example per evidence status. Full run identifiers and reasons are retained to make the classification path inspectable.

State	Example ID	Reason
S1	`official-compat-v1-t3-lo-occ-3-1-region-None-42-20-eae54b2d`	Official-compatible protocol is identifiable, all six folds are complete, and the relative error of each of four disclosed metrics is no greater than 2%.
S2	`audited-t3-lo-occ-3-1-region-None-42-20-efa1ea0d`	Six folds and the main protocol align, but an audited correction is present or at least one metric lies in the 2–5% approximation band.
S3	`audited-t3-ar-occ-3-1-region-None-42-20-3b0793b0`	At least one paper-disclosed metric has relative error above 5%; only trend-level evidence is retained.
S4	`audited-t4-lo-occ-3-1-194-None-42-20-42da10db`	Neither the paper nor repository discloses the Table 4 node identity; node 194 is a predeclared reverse-identification candidate, not a unique truth.
S5	`audited-t3-lstm-occ-6-3-region-None-42-20-c735895e`	The frozen configuration was executed but failed; the failure artifact is retained and cannot contribute a paper number.
S6	`audited-t3-TimesNet-occ-3-3-region-None-42-1-1687eb9d`	The configuration is frozen in the manifest but was not executed or registered after accumulated failures in the same model batch.

A.2 Model identity is a claim field

Names alone do not identify a scientific object. The same family label may refer to an upstream release, a local source snapshot, an existing prediction artifact, a retrained adaptation, or a deterministic composition of other artifacts. Table 3 therefore reports the identity that the frozen artifacts support and names fields that remain unknown. This is especially important for Paris: `Paris_CAPER_phase_only` and `TimeXer_local_audited_compact_L168` are development-only local adaptations, not exact transfers of the UrbanEV experts or claims about every upstream implementation.

A.3 Fail-closed state machine

The portable state sequence is:

$$\texttt{REGISTERED} \rightarrow \texttt{RUNNING} \rightarrow \texttt{CLOSED\_UNVERIFIED} \rightarrow \texttt{AUDIT\_PASS} \rightarrow \texttt{METRIC\_UNLOCKED} \rightarrow \texttt{GATE\_PASS} \text{ or } \texttt{GATE\_FAIL}.$$

Three side transitions prevent a defective artifact from moving forward. First, an incomplete matrix, target/order mismatch, invalid control, boundary violation, or provenance failure moves `CLOSED_UNVERIFIED` to `BLOCKED` or `REPAIR_REQUIRED`; aggregate metrics remain inaccessible. Second, a permitted repair creates `REPAIRED_UNVERIFIED`, records parent and child hashes and whether predictions changed, and returns to the full audit rather than directly to metric unlock. When selective repair could use information about which controls failed, the formal control family is invalidated and rerun non-selectively. Third, `GATE_FAIL` is terminal for the current protocol version. Continuing with a changed model, threshold, loss, or evidence role requires a new versioned protocol and a valid evidence boundary; a failed version is not rewritten as pending.

The two observed incidents instantiate these transitions. In M7-C, the first audit detected three identity permutations at 69/72 cells while `metrics_unlocked=false`; all 24 shuffled controls were invalidated and rerun before the 72/72 audit passed. In Paris, 32 prediction jobs closed without evaluation metrics, missing-target representation was repaired in 24 artifacts (10,236 entries) with predictions unchanged, and the development metrics were unlocked only after the 32/32 prediction audit passed. These are integrity events, not accuracy improvements.

A.4 Portable lock schema and invariants

The project evolved several lock and receipt formats rather than one universal schema. Table 11 gives the normalized interface required to reproduce the paper's claim-control semantics. It is a portable schema, not a statement that every historical artifact contains every field. A missing field must be encoded as `unknown` or fail the relevant gate; it must never be silently inferred from a filename.

Table 11: Normalized lock schema. Existing exemplars use heterogeneous field names; the rightmost column states the invariant a portable implementation must enforce.

Group	Required fields	Existing artifact exemplar	Invariant
Protocol	`protocol_id`, version, status, protocol SHA-256	router, latency, distillation, and Paris teacher protocol files	Data role, endpoint, reference, gate, and stop rule are immutable before aggregate access.
Configuration	`run_id`, config fingerprint, model identity, seed, fold, horizon	status registry and run manifests	One identifier maps to one configuration; retries retain parentage rather than overwriting identity.
Impl.	code/source hashes; dependency and bundle hashes	Chronos model lock; Paris implementation fingerprints	The implementation that emitted predictions equals the registered implementation or has an explicit non-training amendment receipt.
Environment	Python and library versions; accelerator, runtime, precision; optional environment hash	latency hardware/software manifests; Chronos environment block	Scope is explicit. If no unified environment hash exists, reproducibility is limited rather than implied.
Data boundary	source/data hash, role, permitted metadata, fold dates	baseline protocol; Paris boundary lock and source manifest	Development, formal, and protected roles cannot be silently exchanged.
Target identity	target-value, index/order, time, spatial-unit, and mask hashes; fold–horizon cell	expert diagnostics, run status, prediction audits	Candidate and reference use identical target positions and semantics for every compared cell.
Prediction	prediction SHA-256, shape, finiteness/range, attempt path	per-run receipts and prediction audits	The audited bytes are the bytes aggregated; a repair changing predictions requires new scientific evidence.
Metric	metric script/hash, endpoint definition, output hash	metric-lock scripts, aggregate summaries	The metric implementation and weighting rule are fixed; output cannot precede unlock.
Unlock	lock time, conditions, authority, receipt SHA-256, matrix/state hash	Paris metrics-unlock receipt; M7-C formal audit	Every condition is conjunctive and machine-checkable; role-specific unlock does not unlock another role.
Access	analytical-access counter, operation definition, log/receipt location	Paris protected guard and prediction audit	Counter semantics and time origin are stated; storage migration is recorded separately.
Repair	parent hash, child hash, reason, prediction-changed flag, audit return state	shuffled-control amendment and masked-target repair ledger	Repair is traceable and returns to unverified state; non-selective scope is used where observation could bias repair.
Decision	reference, point effect, interval, fold/cell/horizon checks, gate vector, authorized next action	router, Chronos, D1, and Paris teacher summaries/decisions	A failed required component yields terminal failure for that version even when another subcheck passes.
Claim	claim ID, authoritative artifact/hash, evaluation type, allowed/prohibited wording, scope limit	Table 12	Manuscript wording cannot exceed the evidence type or data role.

A.5 Operational definition of analytical access

The Paris guards distinguish three operations.

Storage materialization. Byte-preserving movement, copying, sharding, forensic preservation, and opaque byte hashing of target-bearing files. It also includes creating or verifying a path/size/hash manifest. These operations are recorded as storage events; they do not by themselves authorize or count as target analysis.

Approved metadata handling. Reading the registered non-target fields (date boundary, station identifier, latitude, longitude, postcode, and area), row counts, file sizes, and hashes needed to establish the split and storage receipt, without using state/target values. This is boundary administration, not model evidence.

Analytical access. Any post-lock operation that deserializes, displays, summarizes, plots, filters by, learns from, scores against, or forms a scientific claim from formal/protected target or state values. Manual viewing counts. Feeding values to training, validation, model selection, metric computation, threshold selection, error analysis, or a paper table counts even if no file is modified.

Under this definition, the guard's `analytical_access_count=0` means no protected label was used after the boundary lock for those analytical operations. It does not mean that target bytes were never materialized: one storage-migration event created vault shards and moved recoverable sources. It also does not erase the separately logged display of two example source rows during pre-lock schema discovery. The counter is a procedural project field supported by receipts and audits, not an operating-system reference monitor, append-only security log, encryption guarantee, or defense against a malicious authenticated user.

The in-scope threat is accidental or adaptive analytical use by the project workflow: a script path that reaches a future role, a stale cache, recoverable Git objects in the main project, or a metric opened before the registered closure. The current vault path, manifest, hashes, guards, and negative main-project scan reduce those risks. OS compromise, deliberate bypass by an authorized account, cryptographic confidentiality, and immutable access logging are out of scope. The allowed claim is therefore procedural source separation and zero recorded analytical use, not hardened security.

A.6 Paris fold-synchronous bootstrap, including the partial block

The frozen implementation in `experiments/09_distill_v2/scripts/aggregate_paris_teacher.py` used 5,000 replicates and seed 42. For each development fold it intersected target timestamps across all four horizons, sorted them, required a contiguous hourly sequence, and sliced the sequence at offsets 0, 24, 48, and so on. The final slice was retained even when shorter than 24 hours. Each replicate sampled exactly the fold's number of blocks uniformly with replacement, concatenated the selected timestamp arrays, and reused that same sampled timestamp sequence for every horizon in the fold. Observed-target masks were then applied within each cell, cell RMSEs were macro-averaged, the better CAPER/TimeXer single-model reference was reselected inside the replicate, and teacher gain was computed against that reference. The reported interval is the 2.5th and 97.5th percentiles.

August and October each contain 732 common hourly timestamps, represented as 30 full 24-hour blocks plus one 12-hour block (31 selectable blocks). September and November each contain 708 timestamps, represented as 29 full blocks plus one 12-hour block (30 selectable blocks). The partial block has the same probability of being selected as any other block, but contributes only 12 timestamps whenever selected; consequently replicate timestamp counts can vary with how often the partial block is drawn. No padding, wraparound, discarding, or extra weighting is applied. This exact rule is the scope of the reported Paris interval. No alternative block length or partial-block sensitivity analysis was run for this revision.

A.7 Claim lineage and research-question closure

Table 12 is the manuscript-facing join between a headline result and its machine-readable authority. The hash is computed over the named artifact as stored in the main workspace. A hash identifies bytes; it does not prove that the underlying method is correct. The evaluation-type and wording columns provide the semantic constraint. In particular, the Global Fixed row supports a scoped internal point comparison and a separate measured latency trade-off; it does not acquire an unreported bootstrap interval merely because later failed stages used one.

Table 12: Claim-to-artifact trace and wording guardrails (part 1 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256_{12}	Evaluation type	Allowed wording; prohibited wording
C1	Frozen registry: 1,224 configurations; 865 complete artifacts; S1=156.	`experiments/00_registry_reclassification/outputs/CONFIG_STATUS_SUMMARY.json` `833352c136a4`	registry audit	*Allowed:* In this inventory, completion and exact reproduction differ. *Prohibited:* A field-wide reproducibility rate or a model-quality ranking.
C2	Source-table status distributions are uneven under the frozen mapping.	`experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv` `4b61c2aea510`	registry audit	*Allowed:* Report S1–S6 counts by UrbanEV source table. *Prohibited:* Call an S4 model invalid or intrinsically weak.
C3	Target-dependent CAPER–TimeXer oracle gain is 16.926%.	`experiments/01_expert_diagnostics/outputs/EXPERT_DIAGNOSTIC_SUMMARY.json` `53fa7f22fca6`	real-GT oracle diagnostic	*Allowed:* An infeasible upper bound indicates local complementarity. *Prohibited:* The oracle is deployable or its gain is recoverable.
C4	State-feature expert-win AUC is 0.6151.	`experiments/01_expert_diagnostics/outputs/STATE_FEATURE_DECISION.json` `9f84ac483e02`	self-supervised proxy	*Allowed:* Relative expert advantage is moderately predictable. *Prohibited:* AUC proves lower forecast RMSE or causal demand knowledge.
C5	Strict Global Fixed RMSE is 0.071367, a 3.178% internal gain.	`experiments/03_fixed_fusions/outputs/FIXED_FUSION_SUMMARY.json` `0a0162204c83`	internal rolling real GT	*Allowed:* Improves on TimeXer under the audited UrbanEV protocol. *Prohibited:* Blind-test, cross-city, efficient, or deployment-ready.
C6	Direct-loss router gains 0.443% and fails the frozen 1% gate.	`experiments/05_router_v1/outputs/ROUTER_V1_SUMMARY.json` `a9dfc2ebe4a6`	internal real GT; gated	*Allowed:* A positive sub-threshold internal signal; current stage stopped. *Prohibited:* Router qualification, robustness, or post-hoc gate revision.
C7	Global Fixed batch-1 P50 is 48.86% above TimeXer.	`experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv` `f94d0a7a9e95`	hardware-specific latency	*Allowed:* Measured on the frozen RTX 4060 FP32 sequential workload. *Prohibited:* Portable latency, low cost, or universal serving efficiency.

Table 13: Claim-to-artifact trace and wording guardrails (part 2 of 2). Hashes are the first 12 hexadecimal characters of SHA-256 over the named frozen artifact.

ID	Headline result	Authoritative artifact / SHA-256_{12}	Evaluation type	Allowed wording; prohibited wording
C8	Chronos-2 gains -3.048% versus Global Fixed and fails qualification.	`experiments/06_5_teacher_qualification/outputs/CHRONOS2_QUALIFICATION_SUMMARY.json` `01e8084c16ae`	zero-shot internal screen	*Allowed:* This locked revision does not qualify for this protocol. *Prohibited:* Foundation models are generally inferior for EV data.
C9	Aligned KD gains 0.119% versus identical GT-only; CI crosses zero.	`experiments/07_residual_distillation/outputs/D1_SUMMARY.json` `c38163157c89`	internal real GT; gated	*Allowed:* The primary distillation comparison fails; distill_v1 stops. *Prohibited:* Distillation improves the student or merits D2 latency.
C10	Aligned KD gains 0.259% versus shuffled teacher.	`experiments/07_residual_distillation/outputs/D1_SUMMARY.json` `c38163157c89`	negative control	*Allowed:* Evidence of a small sample-aligned teacher signal. *Prohibited:* Engineering value or a passed primary KD claim.
C11	Paris teacher gains 0.847%, wins 4/4 folds and 16/16 cells, but fails 1%.	`experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json` `5b7c1ddc353c`	development real GT	*Allowed:* Consistent development gain below the frozen qualification effect. *Prohibited:* Round to 1%, imply formal/protected performance, or call it official TimeXer.
C12	Three identity shuffles were blocked at 69/72 before metric unlock.	`experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json` `584313534390`	artifact integrity	*Allowed:* The lock forced a non-selective rerun of all 24 controls. *Prohibited:* Use the invalid controls as performance evidence.
C13	Paris prediction audit passed 32/32 before development-metric unlock.	`experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json` `ee882d8e907b`	artifact integrity	*Allowed:* Identity checks authorized development-metric aggregation. *Prohibited:* The audit proves model correctness or improved RMSE.
C14	Protected analytical-access counter is 0; vault materialization is true.	`experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json` `a7780e39f4ef`	procedural data boundary	*Allowed:* No protected label was used analytically after the boundary lock. *Prohibited:* Targets never existed, were never materialized, or are cryptographically inaccessible.

A.8 Latency across the frozen batch grid

The main text uses batch-one end-to-end P50 as its primary engineering endpoint. Table 14 exposes the complete batch 1/8/32 grid already present in the frozen latency summary. It is generated mechanically by `experiments/10_audit_paper/scripts/generate_revision_appendix.py`; the generator checks for all 18 method–batch rows and formats the existing values without launching the benchmark. The hardware and software context remains the one reported in the original manifests: Windows 10 build 26200, 16 physical/32 logical AMD CPU cores, approximately 16.8 GB RAM, RTX 4060 Laptop GPU with driver 595.71, Python 3.10.8, Torch 2.5.1+cu121, CUDA 12.1, cuDNN 9.1, FP32, and single-threaded Torch/OMP/MKL host settings. GPU clocks were observed per process, not pinned; model loading and checkpoint I/O were excluded. Cold-start, quantized, compiled, parallel-expert, and serverless claims remain outside scope.

Table 14: Frozen end-to-end synchronous latency at batch sizes 1, 8, and 32. P50 and P95 are milliseconds per request; throughput is forecast origins per second at P50. Every value is formatted directly from `LATENCY_SUMMARY_TABLE.csv`; no benchmark was rerun.

Method	Batch origins	E2E P50 (ms)	E2E P95 (ms)	Origins s^{-1} (P50)
DLinear	1	1.166	1.763	857.9
DLinear	8	2.118	2.874	3776.9
DLinear	32	5.382	7.728	5945.7
NLinear	1	0.674	1.056	1483.7
NLinear	8	1.554	2.101	5148.0
NLinear	32	3.896	5.373	8213.7
GBDT-Forecaster	1	6.242	7.195	160.2
GBDT-Forecaster	8	40.188	50.261	199.1
GBDT-Forecaster	32	157.087	186.205	203.7
CAPER	1	7.934	10.798	126.0
CAPER	8	10.740	14.253	744.9
CAPER	32	19.753	24.266	1620.0
TimeXer	1	14.067	15.952	71.1
TimeXer	8	105.780	113.816	75.6
TimeXer	32	482.165	519.827	66.4
Global Fixed CAPER+TimeXer	1	20.940	33.123	47.8
Global Fixed CAPER+TimeXer	8	125.350	136.947	63.8
Global Fixed CAPER+TimeXer	32	479.634	510.574	66.7

Workload: one request produces all four horizons for 275 UrbanEV zones. Measurements used FP32, one RTX 4060 Laptop GPU, sequential expert execution, five processes, 50 warm-ups, and 1,000 timed iterations per process and method–batch cell. Throughput is not monotone across model families because the frozen serving graphs batch differently; these values are workload-specific rather than portable deployment guarantees.

A.9 Protocol and environment artifact index

The following index identifies existing controls; it does not upgrade an incomplete field into a complete reproducibility package. The latency and Chronos stages have explicit environment records. Other stages retain protocol, source/configuration fingerprints, and run receipts but do not expose one uniform cross-stage environment hash, so exact environment equivalence is unknown/not claimed where absent.

Table 15: Protocol and environment artifact index (part 1 of 2).

Artifact	Function	Evidentiary limit
`experiments/00_registry_reclassification/scripts/rebuild_status_registry.py`	Deterministic S1–S6 implementation and priority order	Single programmed classifier; no inter-rater validation
`experiments/00_registry_reclassification/outputs/CONFIG_STATUS.csv`	Row-level run IDs, fingerprints, grades, reasons, and rerun policy	One frozen UrbanEV inventory
`experiments/02_lightweight_baselines/configs/BASELINE_PROTOCOL.json`	UrbanEV target, folds, horizons, seed, and candidate protocol	Candidate-set internal evaluation only
`refine-logs/ROUTER_PROTOCOL_V1_0.md` and `experiments/05_router_v1/configs/ROUTER_V1.json`	Router feature/training configuration and frozen Gate 4	One seed; proxy and real-GT roles remain separate
`experiments/06_static_latency/LATENCY_PROTOCOL_V1_0.md`	Request shape, synchronization, warm-up, repetitions, and primary endpoint	Warm synchronous serving, not cold start
`experiments/06_static_latency/hardware_manifest.json` and `experiments/06_static_latency/software_manifest.json`	Device, driver, platform, runtime, libraries, precision, host threads	One laptop system; clocks observed rather than pinned
`experiments/06_static_latency/LATENCY_SUMMARY_TABLE.csv`	Batch 1/8/32 model-only and end-to-end distributions, throughput, memory	Hardware/workload-specific
`experiments/06_5_teacher_qualification/TEACHER_QUALIFICATION_PROTOCOL_V1_0.md`	Chronos-2 qualification endpoint and gate	Exact project protocol only
`experiments/06_5_teacher_qualification/CHRONOS2_MODEL_LOCK.json`	Resolved model commit, file hashes, snapshot bytes, and environment	One revision and zero-shot profile
`experiments/07_residual_distillation/DISTILL_PROTOCOL_V1_0.md`	Teacher, student, loss, controls, milestones, stop rules	Distill v1 only; later search prohibited after failure
`experiments/07_residual_distillation/configs/FORMAL_STUDENT_CONFIG_V1.json`	338-parameter student input/architecture/training identity	One architecture and seed
`experiments/07_residual_distillation/SHUFFLE_CONTROL_SPEC.json`	24-hour block-shuffle negative-control construction	Control test, not a second primary endpoint

Table 16: Protocol and environment artifact index (part 2 of 2).

Artifact	Function	Evidentiary limit
experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT_V1_0_FAIL_IDENTITY_SHUFFLE.json and experiments/07_residual_distillation/outputs/FORMAL_MATRIX_AUDIT.json	Pre-unlock failure and final 72/72 audit	Supports control integrity, not accuracy
experiments/07_residual_distillation/outputs/D1_SUMMARY.json and experiments/07_residual_distillation/outputs/D1_DECISION.md	Primary/negative-control effects and terminal gate decision	No D2 latency, multi-seed, or protected evidence
experiments/08_paris_new_evidence/DATA_BOUNDARY_LOCK.json and experiments/08_paris_new_evidence/PROTECTED_ACCESS_GUARD.json	Role dates, permitted metadata, storage state, access counter, unlock rule	Procedural guard; not hardened authorization
experiments/08_paris_new_evidence/PARIS_EVIDENCE_VAULT_REPORT.md	Vault identities, hashes, negative main-project scan, threat limits	Metadata/hash audit only; no content-level protected audit
experiments/09_distill_v2/PARIS_TEACHER_QUALIFICATION_PROTOCOL_V1_0.md	Development folds, target, missingness, time rule, teacher fit, gate	Paris development only
experiments/09_distill_v2/PARIS_TEACHER_MODEL_CONFIG.json	Local CAPER/TimeXer profiles, training, queue, bootstrap	Local adaptations, not exact upstream implementations
experiments/09_distill_v2/TIMEXER_LOCAL_ADAPTATION_PROVENANCE.json and experiments/09_distill_v2/PARIS_CAPER_ADAPTATION_V1_0.md	Source hash and explicit local adaptation differences	Upstream TimeXer commit undisclosed; Paris CAPER newly trained
experiments/09_distill_v2/PARIS_TEACHER_DEPLOY_APPROVAL_V1_1.md	Differential review closure after the earlier block	Separate artifact review, not blinded human replication
experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_PREDICTION_AUDIT.json and experiments/09_distill_v2/outputs/teacher_qualification/PARIS_TEACHER_METRICS_UNLOCK_RECEIPT.json	32/32 closure, identities, role-specific metric unlock, access counts	Development-metric authorization only
experiments/09_distill_v2/outputs/teacher_qualification/aggregate/PARIS_TEACHER_QUALIFICATION_SUMMARY.json	Point effect, fold/cell results, synchronized bootstrap, gate vector	Current hypothesis failed; no student/formal/protected result
experiments/09_distill_v2/PARIS_TEACHER_RESULT_REVIEW.md	Independent result-to-claim check and provenance warnings	Warnings limit stronger process claims; they do not reverse the gate
experiments/10_audit_paper/scripts/generate_revision_appendix.py	Mechanical hash/latency table generation	No model, metric, resampling, or vault operation

A.10 Author and reviewer application checklist

Table 17 turns the case-study controls into a review interface. The final row is important: missing validation must be stated as missing. This paper therefore treats the 1% rule as a frozen local convention without an economic utility model, the S1–S6 labels as deterministic without inter-rater evidence, and all model/latency conclusions as bounded to their recorded seeds, panels, adaptations, and serving graph.

Table 17: Portable author and reviewer checklist for auditable forecasting claims.

ID	Check	Author must expose	Reviewer test
1	Target and role semantics	numerator/denominator, spatial unit, mask, sampling clock, and development/formal/protected role	Could two datasets or target constructions be mistaken for the same RMSE scale?
2	Configuration identity	run ID, protocol version, fingerprint, source mapping, seed, fold/horizon, and execution state	Could a completed run still be underidentified or map to several disclosed cells?
3	Information timing	forecast origin, lookback, covariates, fill/scaler fit interval, and prequential weight/tuning source	Does any current/future target influence a parameter evaluated in the same fold?
4	Artifact closure	expected matrix cardinality, terminal attempts, prediction/target/order/time/spatial/mask identities	Are aggregates withheld until every required cell is closed and verified?
5	Metric lock	metric implementation identity, lock state, unlock conditions, authority, receipt, and access counter	Can any result be read or selected before the audit passes?
6	Comparator and endpoint	primary reference, macro/micro weighting, endpoint, sign convention, and secondary diagnostics	Was the reference or endpoint changed after seeing results?
7	Uncertainty and gate	resampling unit, partial-block rule, replicates/seed, minimum effect, consistency checks, stop rule	Is a positive interval against zero being substituted for a failed effect-size requirement?
8	Controls and repair	negative-control construction, invalidation scope, before/after hashes, and whether predictions changed	Were only visibly bad cells repaired, or was the formal control family rerun non-selectively?
9	Model identity	upstream/local source, checkpoint/revision, local adaptations, input contract, and every unknown field	Does a local adaptation masquerade as an official or exact upstream implementation?
10	Engineering measurement	device, software, precision, batch/request shape, synchronization, warm-up, repetitions, P50/P95, throughput	Is a hardware-specific measurement being promoted to portable efficiency?
11	Future/protected evidence	storage versus analytical access, allowed metadata, threat-model limit, logs, and unlock rule	Does the boundary prevent analysis, merely record storage, or claim stronger security than implemented?
12	Claim lineage	claim ID, authoritative artifact, SHA-256, evaluation type, allowed/prohibited wording, scope limit	Can every headline number be traced, and are diagnostics/subchecks kept at their original evidence type?
13	Negative result	terminal decision, disallowed same-interval actions, and requirements for a new version/evidence boundary	Was a failed hypothesis silently continued through tuning, threshold changes, or a new structure?
14	Missing validation	limitations must name absent multi-seed, multi-hardware, external-team, utility, or rater evidence	Is any absent analysis implied by rhetoric such as robust, validated, secure, or general?

Data and artifact availability

All numerical claims in this manuscript map to machine-readable summaries, canonical result tables, protocol locks, and audit receipts in the accompanying experiment directory. Figure and table generation scripts read those artifacts directly. Paris formal and protected target-bearing files are deliberately excluded from the paper artifact and retained only in a procedurally isolated evidence vault; their identities and boundaries are represented by hashes and metadata, not by target values.

Ethics and scope statement

The study uses previously released charging datasets and contains no intervention on human participants. Charging observations may nevertheless encode mobility patterns, so the paper reports only aggregate forecasting evidence and does not attempt individual inference. No protected-test, deployment-readiness, or state-of-the-art claim is made.

References

[1] Han Li, Haohao Qu, Xiaojun Tan, Linlin You, Rui Zhu, and Wenqi Fan. UrbanEV: An open benchmark dataset for urban electric vehicle charging demand prediction. *Scientific Data*, 12(1):523, 2025. doi: 10.1038/s41597-025-04874-4. URL https://doi.org/10.1038/s41597-025-04874-4.

[2] Yvenn Amara-Ouali, Yannig Goude, Nathan Doumèche, Pascal Veyret, Alexis Thomas, Daniel Hebenstreit, Thomas Wedenig, Arthur Satouf, Aymeric Jan, Yannick Deleuze, Paul Berhaut, and Sebastien Treguer. Forecasting electric vehicle charging station occupancy: Smarter mobility data challenge. *Journal of Data-centric Machine Learning Research*, 1(16): 1–27, 2024. URL https://openreview.net/forum?id=0w0wrH5fXQ.

[3] Yvenn Amara-Ouali, Yannig Goude, Nathan Doumèche, Pascal Veyret, Alexis Thomas, Daniel Hebenstreit, Thomas Wedenig, Arthur Satouf, Aymeric Jan, Yannick Deleuze, Paul Berhaut, and Sébastien Treguer. Electric vehicle occupancy of charging points in the city of paris, 2023. URL https://doi.org/10.5281/zenodo.8280566. Dataset.

[4] Zachary J. Lee, Tongxin Li, and Steven H. Low. ACN-Data: Analysis and applications of an open EV charging dataset. In *Proceedings of the Tenth ACM International Conference on Future Energy Systems*, pages 139–149, 2019. doi: 10.1145/3307772.3328313. URL https://doi.org/10.1145/3307772.3328313.

[5] Zihan Guo, Linlin You, Rui Zhu, Yan Zhang, and Chau Yuen. A city-scale and harmonized dataset for global electric vehicle charging demand analysis. *Scientific Data*, 12(1):1254, 2025. doi: 10.1038/s41597-025-05584-7. URL https://doi.org/10.1038/s41597-025-05584-7.

[6] Haohao Qu, Haoxuan Kuang, Qiuxuan Wang, Jun Li, and Linlin You. A physics-informed and attention-based graph learning approach for regional electric vehicle charging demand prediction. *IEEE Transactions on Intelligent Transportation Systems*, 25(10):14284–14297, 2024. doi: 10.1109/TITS.2024.3401850. URL https://doi.org/10.1109/TITS.2024.3401850.

[7] Haohao Qu, Han Li, Linlin You, Rui Zhu, Jinyue Yan, Paolo Santi, Carlo Ratti, and Chau Yuen. ChatEV: Predicting electric vehicle charging demand as natural language processing. *Transportation Research Part D: Transport and Environment*, 136:104470, 2024. doi: 10.1016/j.trd.2024.104470. URL https://doi.org/10.1016/j.trd.2024.104470.

[8] Yuxuan Wang, Haixu Wu, Jiaxiang Dong, Guo Qin, Haoran Zhang, Yong Liu, Yunzhong Qiu, Jianmin Wang, and Mingsheng Long. TimeXer: Empowering transformers for time series forecasting with exogenous variables. In *Advances in Neural Information Processing Systems*, volume 37, pages 469–498, 2024. doi: 10.52202/079017-0015. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/0113ef4642264adc2e6924a3cbbdf532-Abstract-Conference.html.

[9] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(9): 11121–11128, 2023. doi: 10.1609/aaai.v37i9.26317. URL https://doi.org/10.1609/aaai.v37i9.26317.

[10] Abdul Fatir Ansari, Lorenzo Stella, Ali Caner Türkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Bernie Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL https://openreview.net/forum?id=gerNCVqqtR.

[11] Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025. URL https://arxiv.org/abs/2510.15821.

[12] A. P. Dawid. Present position and potential developments: Some personal views: Statistical theory: The prequential approach. *Journal of the Royal Statistical Society. Series A (General)*, 147(2):278–290, 1984. doi: 10.2307/2981683. URL https://doi.org/10.2307/2981683.

[13] Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000. doi: 10.1016/S0169-2070(00)00065-0. URL https://doi.org/10.1016/S0169-2070(00)00065-0.

[14] Hansika Hewamalage, Klaus Ackermann, and Christoph Bergmeir. Forecast evaluation for data scientists: Common pitfalls and best practices. *Data Mining and Knowledge Discovery*, 37(2):788–832, 2023. doi: 10.1007/s10618-022-00894-5. URL https://doi.org/10.1007/s10618-022-00894-5.

[15] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804. URL https://doi.org/10.1016/j.patter.2023.100804.

[16] Joelle Pineau, Philippe Vincent-Lamarre, Koustuv Sinha, Vincent Larivière, Alina Beygelzimer, Florence d'Alché Buc, Emily Fox, and Hugo Larochelle. Improving reproducibility in machine learning research: A report from the NeurIPS 2019 reproducibility program. *Journal of Machine Learning Research*, 22(164):1–20, 2021. URL https://jmlr.org/papers/v22/20-303.html.

[17] Yvenn Amara-Ouali, Yannig Goude, Pascal Massart, Jean-Michel Poggi, and Hui Yan. A review of electric vehicle load open data and models. *Energies*, 14(8):2233, 2021. doi: 10.3390/en14082233. URL https://doi.org/10.3390/en14082233.

[18] Mamunur Rashid, Tarek Elfouly, and Nan Chen. A comprehensive survey of electric vehicle charging demand forecasting techniques. *IEEE Open Journal of Vehicular Technology*, 5: 1348–1373, 2024. doi: 10.1109/OJVT.2024.3457499. URL https://doi.org/10.1109/OJVT.2024.3457499.

[19] Luboš Buzna, Pasquale De Falco, Gabriella Ferruzzi, Shahab Khormali, Daniela Proto, Nazir Refa, Milan Straka, and Gijs van der Poel. An ensemble methodology for hierarchical probabilistic electric vehicle load forecasting at regular charging stations. *Applied Energy*, 283:116337, 2021. doi: 10.1016/j.apenergy.2020.116337. URL https://doi.org/10.1016/j.apenergy.2020.116337.

[20] Christoph Bergmeir and José M. Benítez. On the use of cross-validation for time series predictor evaluation. *Information Sciences*, 191:192–213, 2012. doi: 10.1016/j.ins.2011.12.028. URL https://doi.org/10.1016/j.ins.2011.12.028.

[21] Vitor Cerqueira, Luis Torgo, and Igor Mozetič. Evaluating time series forecasting models: An empirical study on performance estimation methods. *Machine Learning*, 109(11):1997–2028, 2020. doi: 10.1007/s10994-020-05910-7. URL https://doi.org/10.1007/s10994-020-05910-7.

[22] Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001. URL https://doi.org/10.1016/j.ijforecast.2006.03.001.

[23] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4): 1346–1364, 2022. doi: 10.1016/j.ijforecast.2021.11.013. URL https://doi.org/10.1016/j.ijforecast.2021.11.013.

[24] Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Nazanin Mohammadi Sepahvand, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks. In *Proceedings of Machine Learning and Systems*, volume 3, pages 747–769, 2021. URL https://proceedings.mlsys.org/paper_files/paper/2021/hash/0184b0cd3cfb185989f858a1d9f5c1eb-Abstract.html.

[25] Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. URL https://www.jmlr.org/papers/v11/cawley10a.html.

[26] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349 (6248):636–638, 2015. doi: 10.1126/science.aaa9375. URL https://doi.org/10.1126/science.aaa9375.